

UPSC Prelims

Environment & Ecology

Class No. 12

Sustainable Development

Renewable Energy

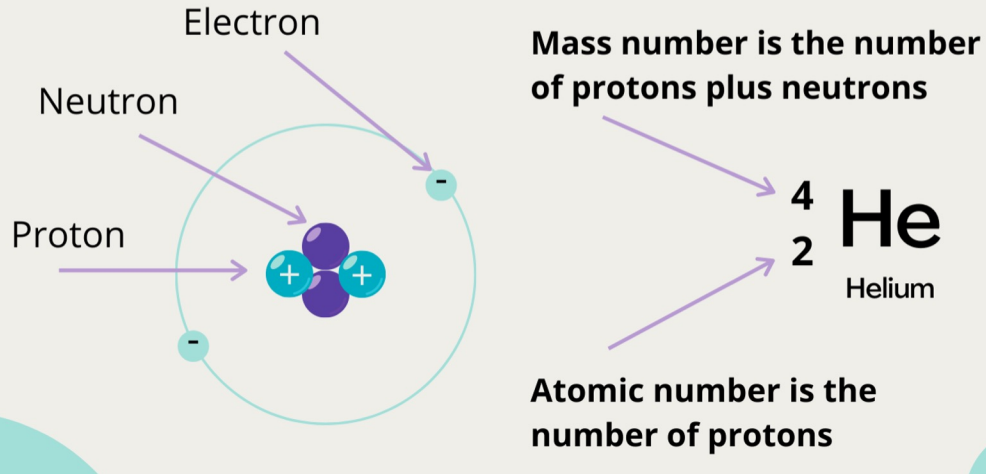


- **Nuclear Energy**
- Generation of energy from nuclear reaction
- Can be through
 1. Nuclear Fission
 2. Nuclear Fusion
- Nuclear fission involves splitting of heavy nuclei of uranium, thorium or plutonium (also called nuclear fuels) into lighter daughter nuclei.

Renewable Energy

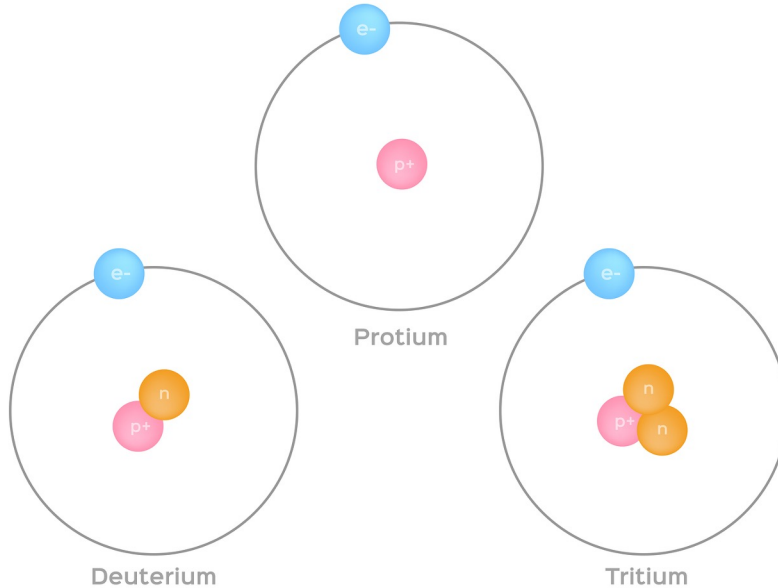


ATOMIC NUMBER VS MASS NUMBER

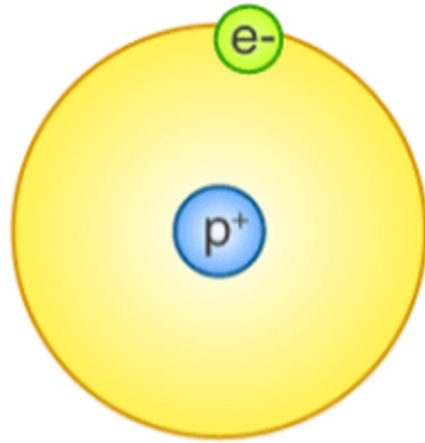


Concept of Isotopes, electrons, protons

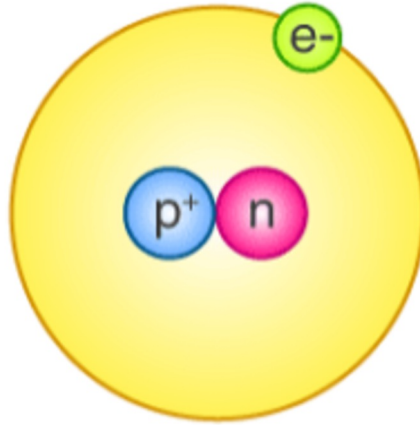
Hydrogen Isotopes



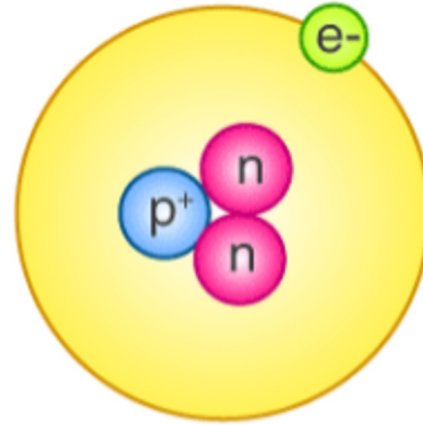
Concept of Isotopes, electrons, protons



Hydrogen-1
Mass number : 1



Hydrogen-2
Deuterium
Mass number : 2



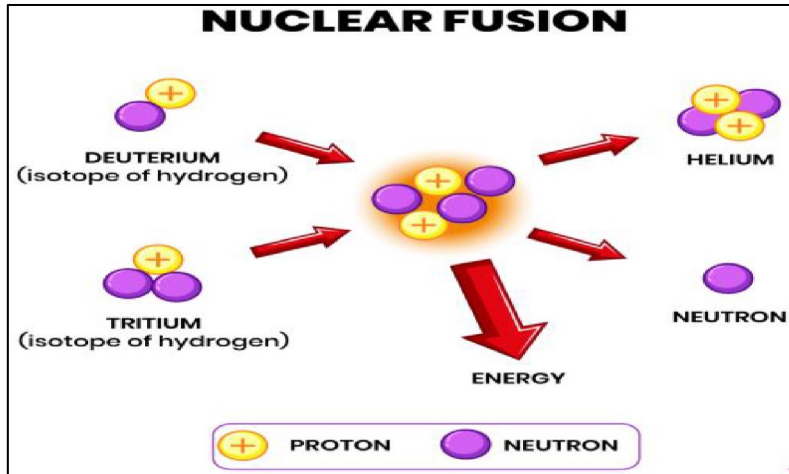
Hydrogen-3
Tritium
Mass number : 3

Renewable Energy

Difference between Nuclear fission or fusion

Criteria	Nuclear Fission
Energy release through	Splitting of a heavy, unstable nucleus into two lighter nuclei
Process	 The diagram illustrates the process of nuclear fission. On the left, a yellow sphere labeled 'Neutron' is shown moving towards a large, grey, spherical nucleus labeled 'Uranium-235'. Upon impact, the nucleus splits into two smaller, grey, spherical nuclei labeled 'Lighter Element'. A large yellow starburst labeled 'Energy' is shown at the point of impact. To the right, a yellow sphere labeled 'Neutron' is shown moving away from the split nucleus. The entire process is labeled 'Nuclear Fission' at the bottom.
Fuel	Uranium and plutonium
Amount of energy produced	Lesser than that in nuclear fusion
Waste generated	Highly radioactive fission products
Operational life	Additional neutrons released can initiate a Chain reaction sustaining fission reactions for longer durations

- **Nuclear Fusion**
- Process by which **two light atomic nuclei** (for example tritium and deuterium) combine to form a single heavier (Helium) one while releasing massive amounts of energy.



Renewable Energy

Nuclear Fission vs. Nuclear Fusion

> Nuclear fission is the **splitting of a nucleus** into smaller particles, releasing high amount of energy.



> **Not common** in nature.



> May require **high-speed neutrons**.



> Produce a **high energy**.



> **Examples:** neutron bombardment of uranium-235 and radioactive decay in unstable isotopes.



> Nuclear fusion is the **combination of two smaller atoms** to create a large atom releasing energy.



> **Common** in stars such as the Sun.



> Require **high temperature and high pressure** conditions.



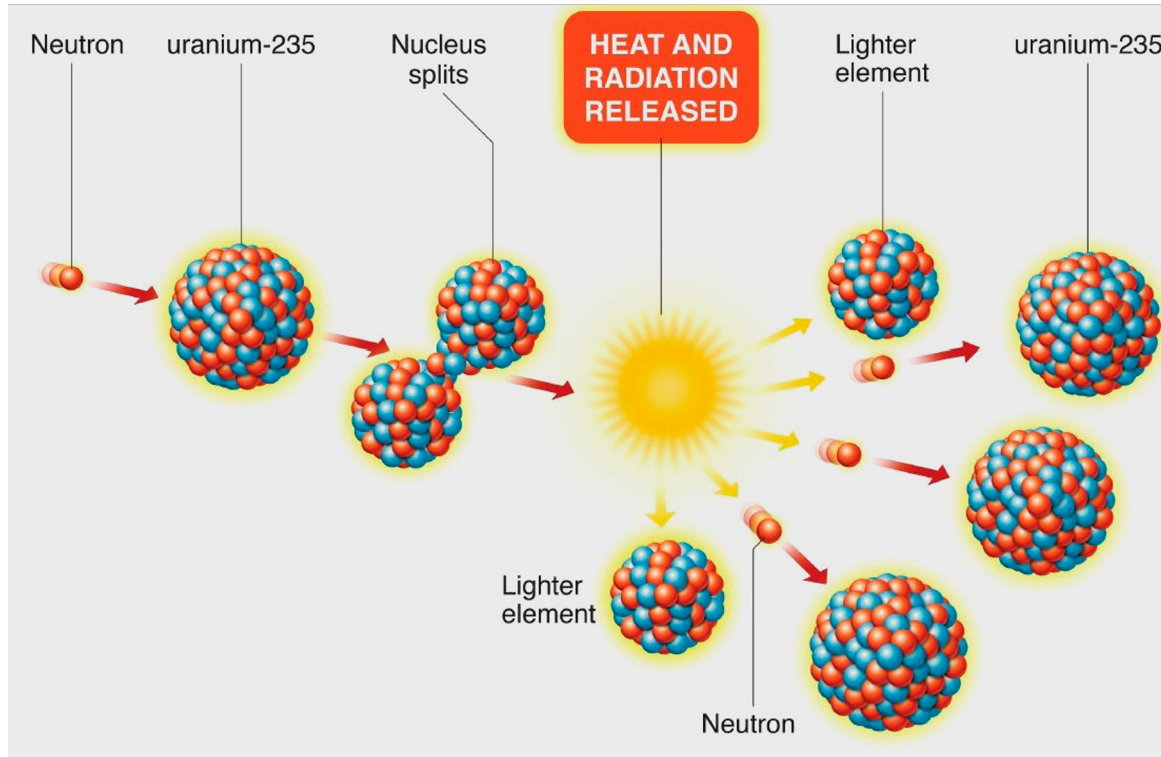
> Reactions of **light nuclei** produce a **very high energy**; reactions of **heavy nuclei** may not release energy.



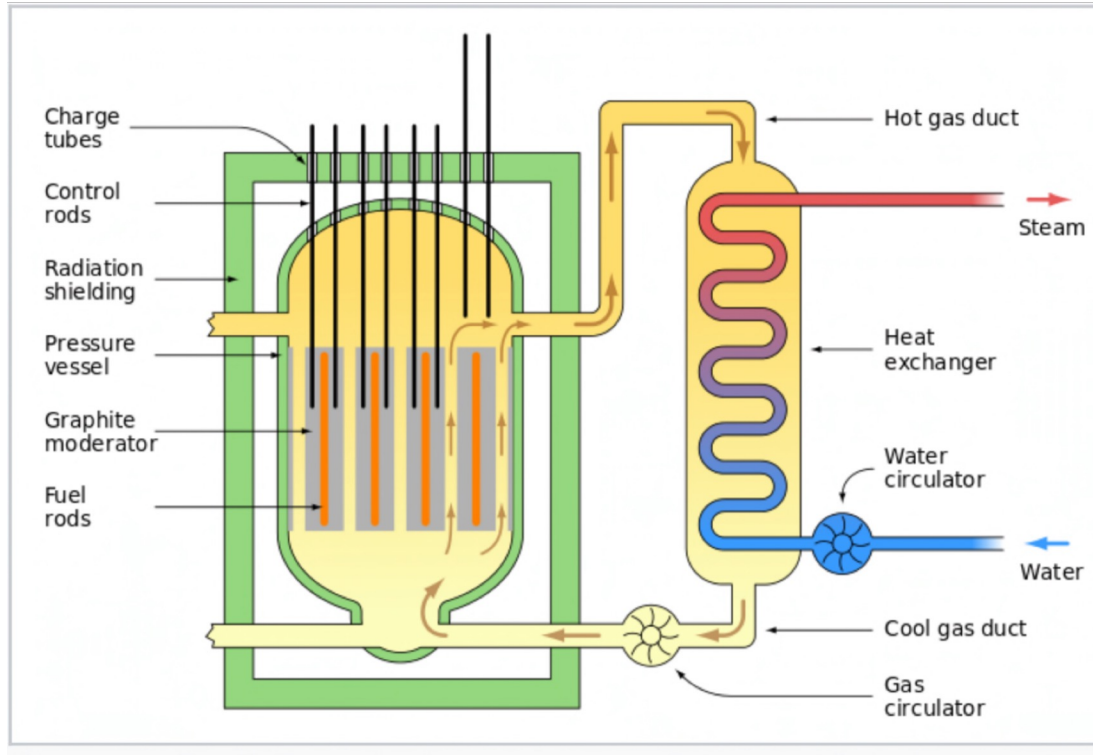
> **Examples:** fusion between Deuterium and Tritium.



Nuclear Fission



Nuclear Fission



Nuclear fusion

- Nuclear reactor is based on **controlled chain reaction** based on fusion
- A **neutron moderator** is a medium that reduces the speed of fast neutrons, thereby turning them into thermal neutrons capable of sustaining a nuclear chain reaction.
- Eg- regular (light) water (in 74.8% of the world's reactors), solid graphite (20% of reactors), heavy water (5% of reactors)
- **Coolant**- A nuclear reactor coolant — usually water or molten salt — is circulated past the reactor core to absorb the heat that it generates.
- The **heat** is carried away from the reactor and is then used to **generate steam**.

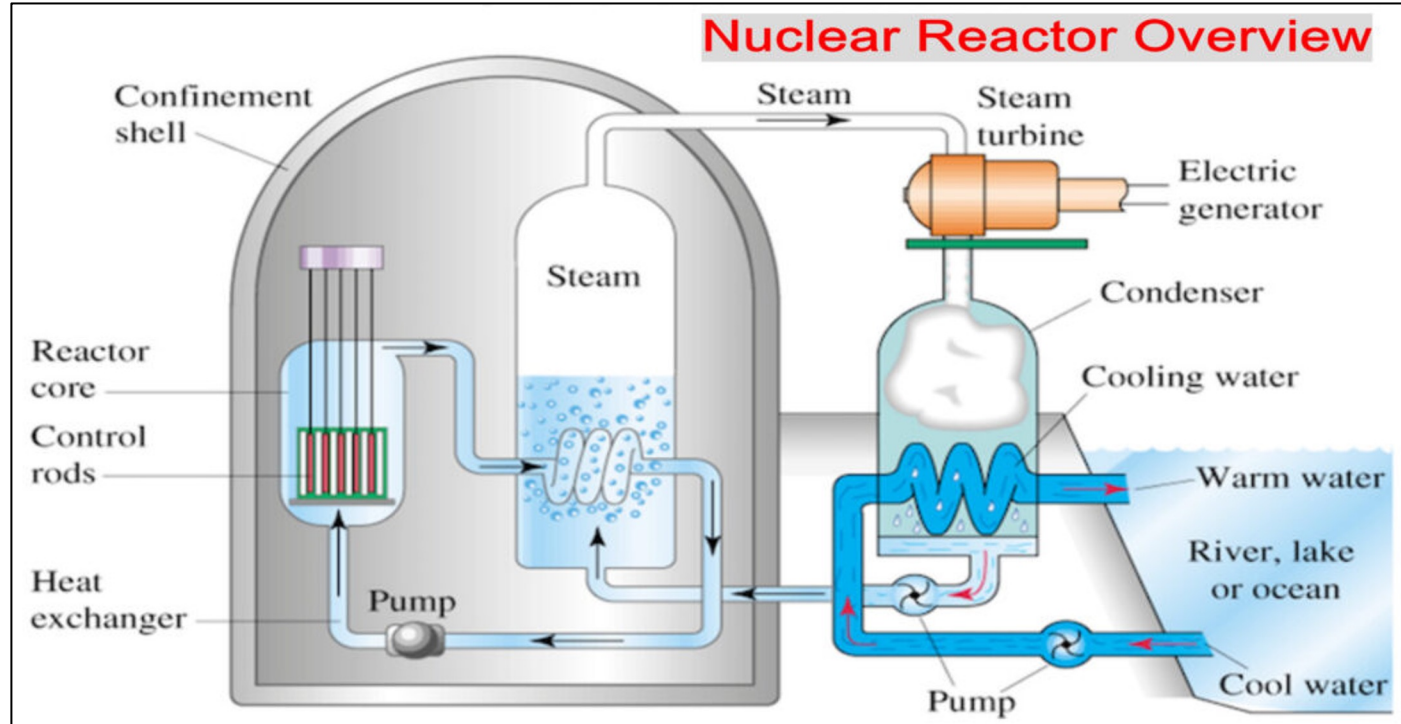
- **Control Rods or Reactivity control-** The power output of the reactor is adjusted by controlling how many neutrons are able to create more fissions. Control rods that are made of a neutron poison are used to absorb neutrons.
- Absorbing more neutrons in a control rod means that there are fewer neutrons available to cause fission.
- So pushing the **control rod deeper** into the reactor will **reduce its power output**, and extracting the control rod will increase it.
- Control rods are composed of chemical elements such as **boron**, silver, indium and cadmium.

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- Moderators slow down neutrons
- Control Rods absorb neutrons
- Moderators are like accelerators
- Control Rods are like brakes

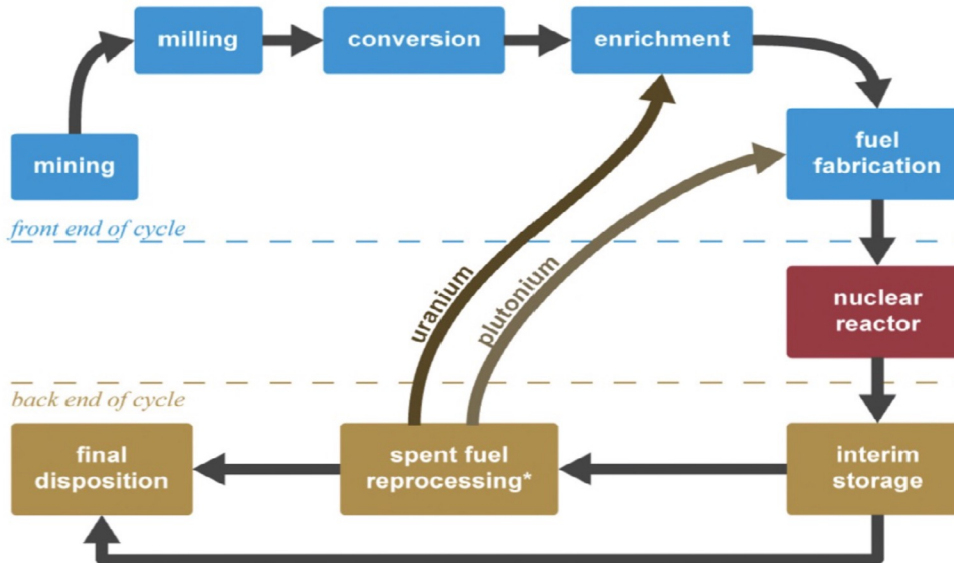
Renewable Energy



- **Nuclear fuel cycle**
- Refers to operations and facilities required for nuclear power generation
- Divided into 2 parts-
 1. Front end (before generation)
 2. Back end (after generation)

Renewable Energy

Nuclear fuel cycle



*Spent fuel reprocessing is omitted from the cycle in most countries, including the United States.

- **Front end**

1. **Survey and exploration-** Atomic Minerals directorate (AMD) Hyderabad

2. **Mining and Milling** UCIL (Uranium Corporation of India Ltd.) located at Jaduguda, Singhbhum, Jharkhand [New deposits also found in Tummalapalle, Andhra Pradesh]

- New uranium sites (usually found in plateaus)
- Domiasiat- Meghalaya
- Rohil- Rajasthan
- Gwalior Basin in Madhya Pradesh

3. Enrichment

- Natural uranium is only **0.7% U-235**, the fissionable isotope.
- The other **99.3%** is **U-238** which is not fissionable.
- The uranium is usually enriched to 2.5-3.5% U-235 for use in light water reactors.
- The enriched uranium fuel used in fission reactors cannot be used to make a bomb.
- It takes enrichment to over **90%** to obtain the fast chain reaction necessary for **weapons applications**.
- Enrichment to 15-30% is typical for breeder reactors.
- Done by UCIL

4. Fabrication

- U 235 and U 238 converted to oxide or carbide **solid yellow cakes** and made into pellets.
- These pellets are packed into fuel rods. (Nuclear fuel complex, Hyderabad)

4. Power Generation

- **Back end**

1. **Reprocessing**- done by Bhabha Atomic Research Centre (BARC)

Unused uranium is recovered and radioisotopes are removed.

Unused uranium : enrichment and fabrication.

Nuclear waste : waste management.

- **Back end**

2. **Nuclear waste management : done by BARC**

- Nuclear waste is **vitrified** (solidified) and stored for some time and dumped into deep geological formations
- Eg- abandoned mines, oceans, rocks

- **Enrichment and Reprocessing**
- Country that has achieved full cycle has established front and backend process. Country with ENR technology is nuclear capable and develop dual use nuclear technology (weapons and peaceful uses)
- **International Atomic Energy Agency**
- Nuclear watchdog of UN
- Headquarters- Vienna, Austria
- **Role**
- Peaceful use of nuclear tech

Renewable Energy



- Nuclear security (to ensure it does not go to irresponsible actors)
- Nuclear safety: regulations etc
- **Nuclear safeguards:** peaceful should not be diverted to military purpose.
Eg- NPT, CTBT
- The nuclear facilities are kept under IAEA safeguards if the source of Uranium, which is the fissile material for a nuclear reactor, is from outside the territory of India or if the new reactor plants are established with foreign collaboration.
- This is to ensure that imported uranium was not diverted for military use and to assure that the imported uranium is used to generate nuclear energy for civilian purposes.
- Relate with Nuclear Supplier's group : is india a member?

Renewable Energy



International Atomic Energy Agency (IAEA)



About: Set up as world's **Atoms for Peace** organization in 1957 within United Nations, IAEA is the **international centre for cooperation in the nuclear field**.



Membership: 175 member states. India is founding member.



Mandate:

- ➔ **Maximize contribution of nuclear technology to society** while verifying its peaceful use.
- ➔ **Promotes exchange of scientific and technical information** among Member States.



Other key information:

- ➔ **Policy-making bodies** : Board of Governors and annual General Conference of IAEA Member States.
- ➔ **Under Non-Proliferation of Nuclear Weapons (NPT)**, IAEA has a specific role as **international safeguards inspectorate**.

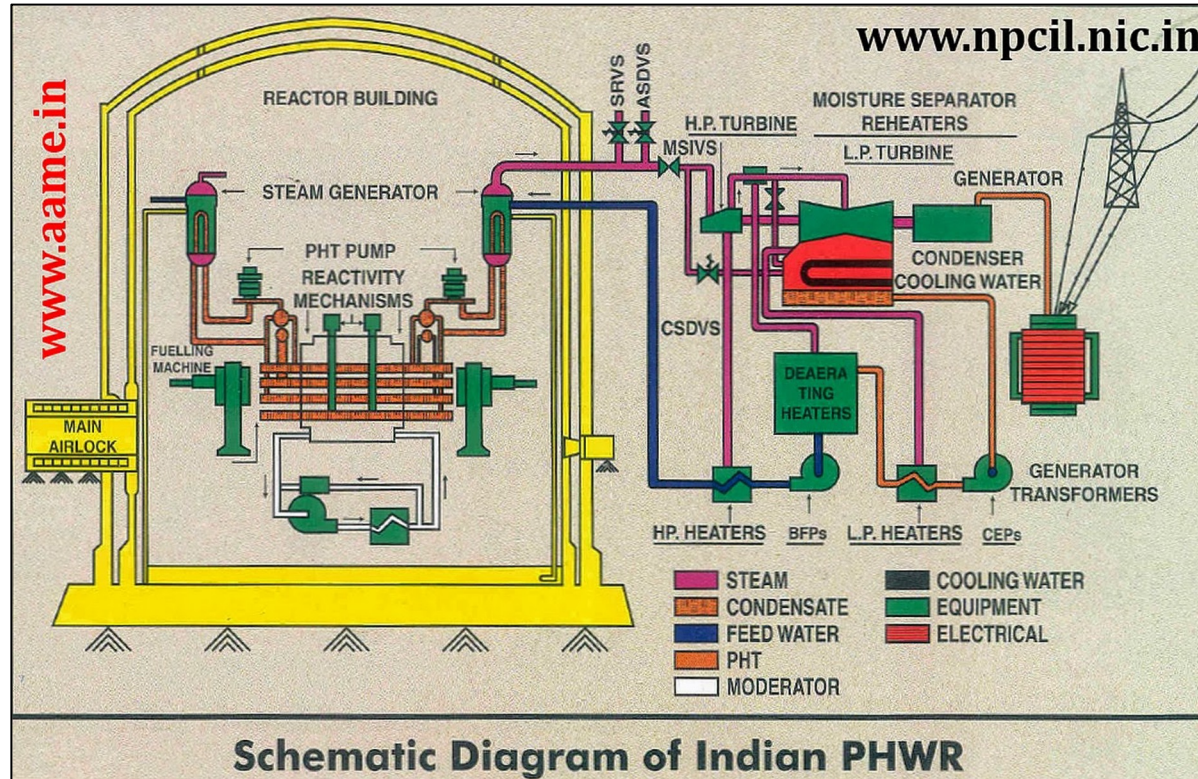


- **Types of nuclear reactors**
- **Boiling water reactor**
- 1st generation reactor which uses normal water to boil to steam in core of reactor which produces a mixture of water-steam
- Steam is separated to run turbine generator and produce electricity
- Steam is converted to liquid water and heated again
- Example- Fukushima nuclear plant

- **Advantages**
- No heat exchange, direct water conversion > **more power**
- Simple design and operation
- **Disadvantages**
- Water in **core** is **radioactive**. Water leakage will contaminate
- Difficult to control risks due to leakage
- **Fukushima earthquake** > generator circulating water stopped > hot water stagnation and explosion

- **Pressurized heavy water reactor (PHWR)**
- Used in India : India is largest producer of D₂O
- It uses **heavy water (D₂O)** both as moderator and coolant passing through core of reactor in 1st cycle
- Heavy water becomes super hot due to **high pressure** without converting to steam.
- This superhot heavy water exchange heat to 2nd cycle containing normal water which gets converted to steam to run turbine and produce electricity

Renewable Energy

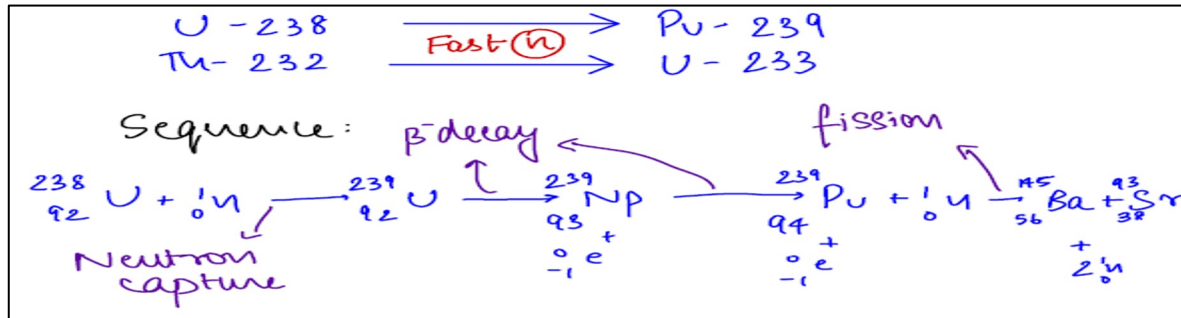


Renewable Energy



- **Advantages**
- **Safer** as **no steam** in core, Steam can be condensed and recycled
- Higher efficiency
- **Disadvantages**
- **Costly** as it used Heavy water in place of water in core
- Complex design and procedure

- **Fast Breeder Reactor**
- It uses fast neutron to convert **U 238 into Pu 239** (fissile)
- Generated more amount fuel than it consumes
- Coolant is **liquid sodium**
- No moderator is required as no need to slow down the neutron



- **Advantages**
- No use of heavy water so **lower costs** (liquid sodium is expensive though)
- Better use of **U-238** which is **more abundant** in earth crust and recovered from reprocessed spent fuel of BWR and PHWR
- Pu-239 generated can be reprocessed and can be used in making Nuclear-weapons
- **Disadvantages**
- More control is required as **Sodium is highly flammable** and any leakage may cause disaster
- No use of heavy water so difficult to control

- **Thorium Reactor: Advanced Heavy Water Reactor (AHWR)**
- **Combination** of tech of BWR and PHWR
- Used to convert normal water into steam and used as coolant whereas heavy water is used as moderator
- Fuel: Thorium- Uranium-233 mixture oxide
- **Advantages**
- Thorium is **more abundant** than Uranium (**Monazite sands**) [4 times more abundant in earth's crust]
- Produces **less radioactive waste** than BWR and PHWR.

Renewable Energy



- **Disadvantages**
- Since water is converted into steam directly in the core, **safety** is required
- Consumes **more water** as water used is radioactive.

- **India's nuclear program**
- Context- Energy and security needs and utilize **thorium reserves (25%)**
- Conceptualized in 1954 by **Homi J Bhabha** and subsequently Atomic Energy Act was passed in 1962
- However, Thorium not fissile and needs to be converted to U-233 which requires Pu-239.
- This is done using **3-stage nuclear program**

Renewable Energy

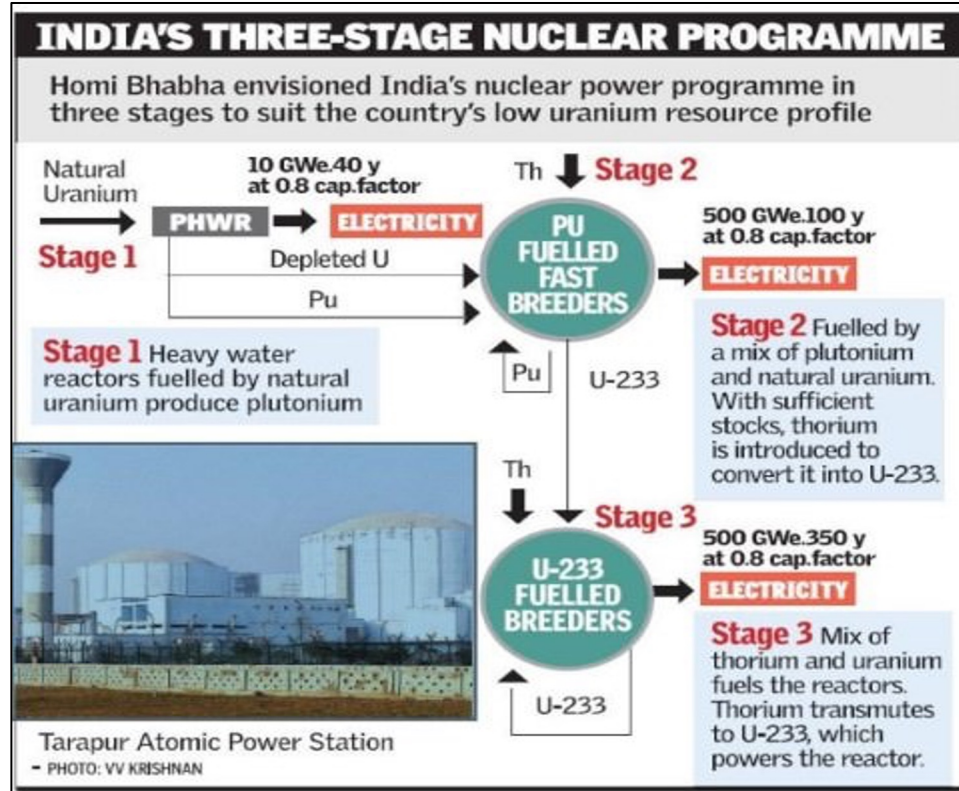


- **First stage**
- First Stage : BWR /PHWR
- **Natural Uranium** as fuel, heavy water as both moderator and coolant
- Plutonium output : used in stage 2
- **Most** reactors are in **1st Stage**

- **Second stage**
- **Fast Breeder Reactor** : Breeds more fuel than it consumes
- Moderators are not required : as fast neutron is what is needed
- Raw material : **Pu- 239**
- It experiences fission, produces energy, U-238 converts to 2 extra PU-239
- After building Pu-239 inventory : thorium is used as a blanket material and transformed to uranium – 233
- Then Thorium can be used in further stages
- India's first prototype fast breeder reactor : **Kalpakkam (Tamil Nadu)**
- Design : Indira Gandhi Center for Atomic Research
- Operator : Bhartiya Nabhikiya Vidyut Nigam Limited (Bhawini)

- **Third Stage**
- **Advanced Heavy water reactor**
- Currently in experimental stage
- Moderator : light water
- Coolant : Heavy Water- D₂O
- **Future programme**

Renewable Energy

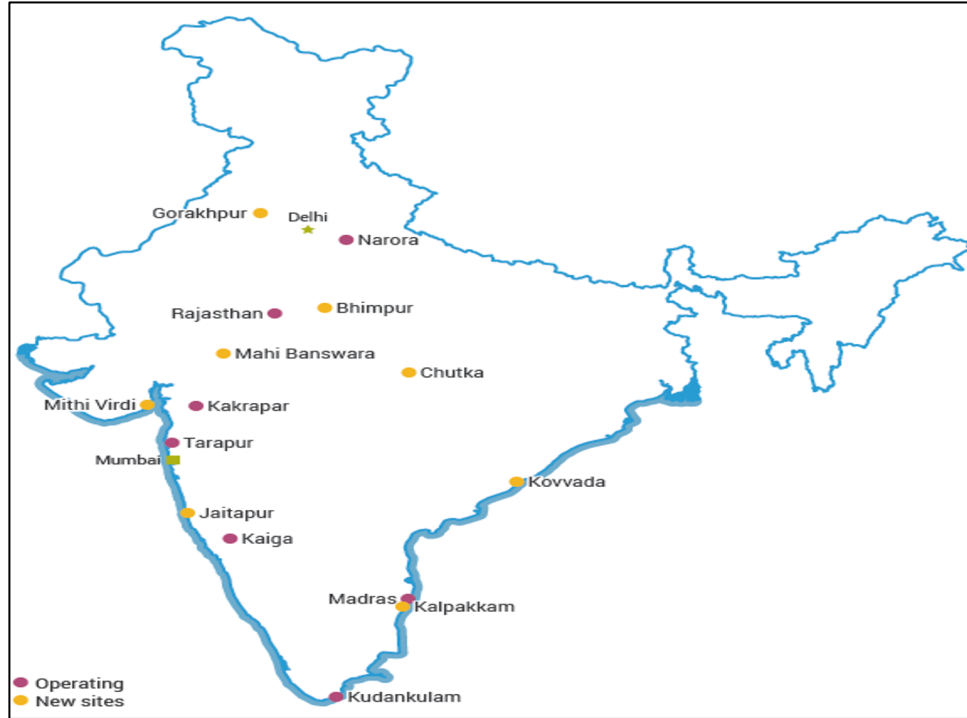


Renewable Energy



Nuclear Power plants in Operation	Nuclear Power Plants under Construction	Planned Nuclear Power Plants
▪ Rawatbhata (Rajasthan) ▪ Tarapur (Maharashtra) ▪ Kudankulam (Tamil Nadu) ▪ Kakrapar (Gujarat) ▪ Kalpakkam (Tamil Nadu) ▪ Narora (Uttar Pradesh) ▪ Kaiga (Karnataka)	▪ Kakrapar 3&4 (Gujarat) ▪ Rawatbhata (Rajasthan) ▪ Kudankulam 3&4 (Tamil Nadu) ▪ Kalpakkam PFBR (Tamil Nadu)	▪ Jaitapur (Maharashtra) ▪ Kovvada (Andhra Pradesh) ▪ Mithi Virdi (Gujarat) ▪ Haripur (West Bengal) ▪ Gorakhpur (Haryana) ▪ Bhimpur (Madhya Pradesh) ▪ Mahi Banswara (Rajasthan) ▪ Kaiga (Karnataka) ▪ Chutka (Madhya Pradesh) ▪ Tarapur (Maharashtra)

Renewable Energy



- **India's Nuclear Doctrine**
- **No first use**
- No use against non nuclear state
- Credible minimum deterrence
- **Civilian control** of nuclear weapon
- Can use as retaliation to **biological and chemical weapon**
- Unacceptable damage in case of first strike

- **Nuclear Fusion**
- Process by which **two light atomic nuclei** (for example tritium and deuterium) combine to form a single heavier (**Helium**) one while releasing massive amounts of energy.
- Fusion reactions take place in a state of matter called **plasma** a hot, charged gas made of positive ions and free-moving electrons with unique properties distinct from solids, liquids, or gases.
- **Challenge to fusion:**
 - Strong **repulsive electrostatic forces** between the positively charged nuclei prevent them from getting close enough together to collide and for fusion to occur.

- **When does it happen:** If the **attractive nuclear force** (which binds protons and neutrons together in atomic nuclei) between the nuclei **outweighs** the **repulsive (electrostatic) force**, then a fusion reaction can occur.
- Such conditions can occur when the temperature increases, causing the ions to move faster and eventually reach speeds high enough to bring the ions close enough together.
- Conditions required to achieve fusion
 - Temperature of more than **100 million degrees Celsius**.
 - Maintaining a high enough density for a long enough time so that the rate of fusion reactions will be large enough to generate the desired power.

- Types of Nuclear Fusion: Inertial and Magnetic:

Inertial	Magnetic
Laser or ion beams are focused very precisely onto the surface of a target. This results in very high temperatures.	hundreds of cubic meters of plasma at a density of less than a milligram per cubic meter are confined by a magnetic field at high pressure and heated to fusion temperature.
It is relatively easier to attain break-even energy levels through inertial fusion compared to magnetic fusion.	

- Advantages of fusion: abundant energy, fuels are widely available, don't emit harmful gases like CO₂, much **less radioactive waste** compared to fission etc.
- **Challenges in fusion process**
- At such high temperatures, matter exists only in plasma state, which is extremely difficult to work with.
- Fusion reactions **are not easily controlled** due to high temperatures.
- It was in news because In a fusion experiment using lasers, the scientists at Lawrence Livermore National Laboratory in California achieved a net energy gain for the first time

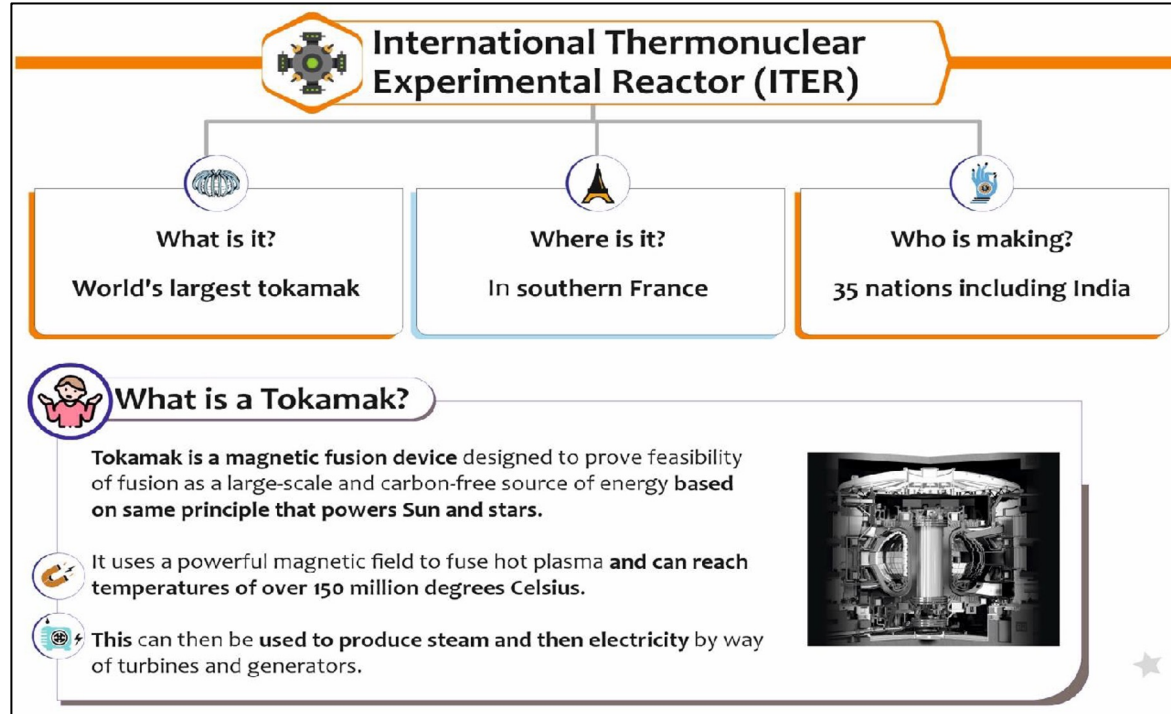
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- **Achieving Ignition:** Scientists for the first time ever, were able to achieve ignition in nuclear fusion i.e., creating a nuclear reaction that generates more energy than it consumes.
- **Execution:** Used powerful lasers to heat and compress hydrogen nuclei. When the nuclei fuse, they release heat. When this heat is equal to or greater than the heat delivered to the container, the event is called ignition.

- **Initiatives to harness nuclear fusion energy**
- **International Thermonuclear Experimental Reactor (ITER)** Assembly.
- India has supplied **Cryolines** for ITER project in France.
- **China's Artificial Sun:** The Experimental Advanced Superconducting Tokamak (EAST) device replicates the nuclear fusion.
- SST-2 tokamak: India's experimental fusion reactor at the Institute of Plasma Research in Gujarat.
- The Joint European Torus (JET): project for opening the way to future nuclear fusion grid energy.

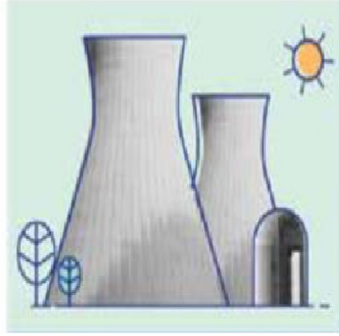
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- **Recent experiment**
- In a fusion experiment using lasers, the scientists at **Lawrence Livermore National Laboratory** in California achieved a **net energy gain** for the first time.
- **Achieving Ignition:** Scientists for the first time ever, were able to achieve **ignition** in nuclear fusion i.e., creating a nuclear reaction that generates more energy than it consumes.
- **Execution:** Used powerful lasers to heat and compress hydrogen nuclei. When the nuclei fuse, they release heat. When this heat is equal to or greater than the heat delivered to the container, the event is called ignition.

- **Small modular reactors (SMRs)**
- SMRs are **advanced nuclear reactors** that have a power capacity of up to 300 MW(e) per unit.
- According to the International Atomic Energy Agency (IAEA), more than 70 SMR concepts are currently under development in 18 countries.
- **Advantages of SMRs**
- **Smaller footprint:** SMRs can be sited on locations not suitable for larger nuclear power plants.
- **Affordable:** Prefabricated units of SMRs can be manufactured and then shipped and installed on site.
- Savings: SMRs offer savings in **cost** and **construction time**

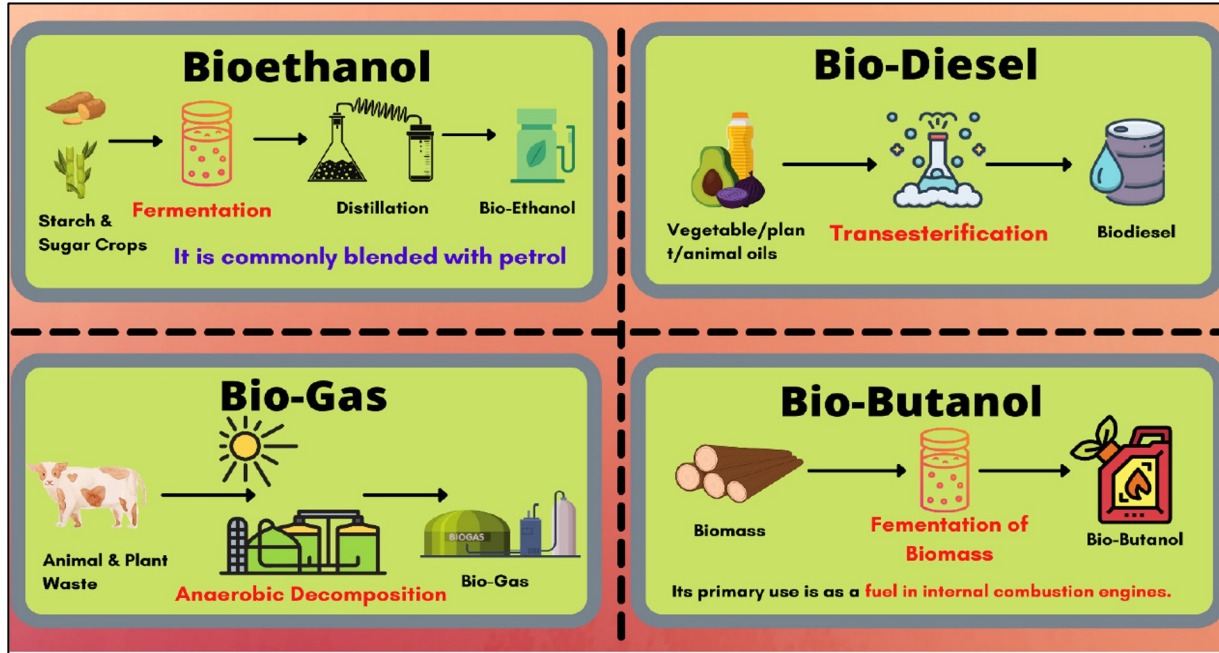
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	Microreactors	Small Modular Reactors	Conventional Nuclear Reactors
			
Power Capacity	Fewer than 10 MW (e) (Megawatt electric)	Up to 300 MW (e)	700+ MW (e)
Refueling	Years without refueling	Every 3 to 7 years	1 and 2 years ★

- **Biofuels**
- Any hydrocarbon fuel i.e., solid, liquid or gas, that is produced from an organic matter which may be living or once living material, in a short period of time is considered a biofuel.
- Examples of biofuels include:
 - Solid: Wood, manure
 - Liquid: Bioethanol and Biodiesel
 - Gaseous: Biogas
- As biofuels emit less carbon dioxide (CO₂) than conventional fuels they can be blended with existing fuels as an effective way of reducing CO₂ emissions in the transport sector.

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- Important Biofuels

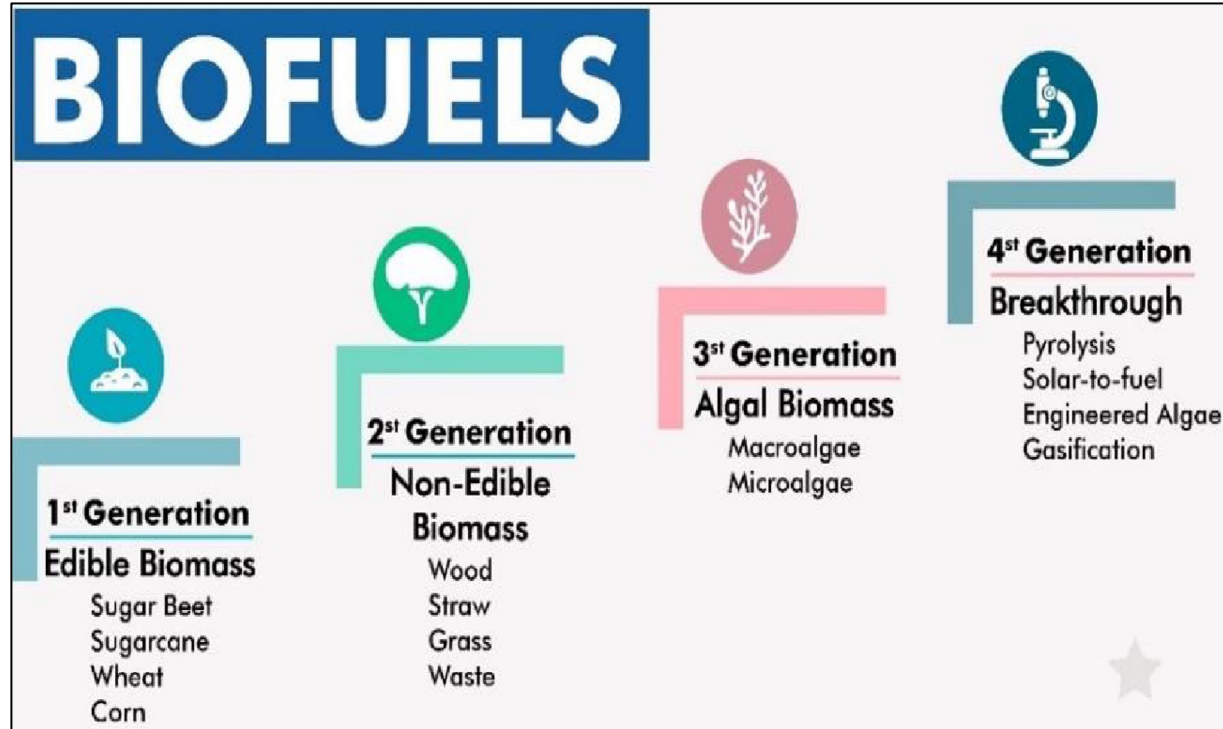


- **Generations of Biofuels**
- **First Generation Biofuels (1G)**
- The First Generation biofuels are produced directly from **food crops** by abstracting the oils for use in biodiesel or producing bioethanol through conventional methods like fermentation.
- Crops such as **wheat** and **sugar** are the most widely used feedstock for **bioethanol** while **rapeseed oil** has proved highly effective for use in **biodiesel**.
- **Issues**: These biofuels can produce **Negative Net** energy gains, releasing more carbon in their production than their feedstock's capture in their growth. The most contentious issue is **'fuel vs food'**. Biofuels from foodgrains has been blamed for the increase in food prices over the last couple of years

- **Second Generation Biofuels (2G)**
- The 2G biofuels are produced from **non-food crops** such as **wood, organic waste**, food crop waste & specific biomass crops, therefore eliminating the main problem with first generation biofuels.
- They are also aimed at being more cost competitive in relation to existing fossil fuel.
- Life cycle assessments of second-generation biofuels have also indicated that they will increase **Positive net energy gains**, overcoming another of the main limitations of first-generation biofuels.

- **Third Generation Biofuels (3G)**
- The Third Generation of biofuels takes advantage of **specially engineered energy** crops such as **algae**.
- The algae are cultured to act as a low-cost, high-energy, and entirely renewable feedstock.
- It is predicted that algae will have the potential to **produce more energy per acre** than conventional crops.
- Algae can also be grown using land and water unsuitable for food production. A further benefit of algae based biofuels is that the fuel can be manufactured into a wide range of fuels such as diesel, petrol, and jet fuel.
- It is potentially **carbon neutral** (the same amount of carbon is absorbed and emitted).

- **Fourth Generation Biofuel (4G)**
- The Fourth Generation biofuels are aimed at not only producing sustainable energy but also a way of **capturing and storing CO₂**. Biomass materials, which have absorbed CO₂ while growing, are converted into fuel using the same processes as second-generation biofuels. This process differs from second and third generation production as at all stages of production the carbon dioxide is captured using processes such as **oxy-fuel combustion**.
- The carbon dioxide can then be **geo-sequestered** by storing it in old oil and gas fields or saline aquifers.
- This carbon capture makes 4th generation biofuel production **carbon negative** rather than simply carbon neutral, as it locks away more carbon than it produces. This system not only captures and stores carbon dioxide from the atmosphere, but it also reduces CO₂ emissions by replacing fossil fuels.



- **Advantages of Biofuels**
- Increases life of Vehicle Engine (They have higher cetane & better lubricating properties)
- **Less** carbon emissions
- **Easy to source-** Biofuels are made from many different renewable sources such as manure, waste from crops, corn, switchgrass, soybeans, algae and plants grown specifically for the fuel.
- **Lower Levels of Pollution-** Biofuels are biodegradable that reduces the possibility of soil contamination and contamination of underground water during transportation, storage or use.

- **Disadvantages of Biofuels**
- High Cost of Production and Future Price- Biofuels are quite expensive to produce in the current market
- Food vs fuel debate
- Pollution- Large scale industries meant for churning out biofuel are known to emit large amounts of emissions and cause small scale water pollution as well.
- Changes in Land Use and Pollution- Biofuel production can encourage monoculture.
- Biofuel is less suitable for use in low temperatures- attract moisture + increases microbial growth in the engine that clogs the engine filters.

- **National Policy on Biofuels (2018)**
- National Policy on Biofuel 2018 is a revised version of National Policy on Biofuels made by Ministry of New and Renewable Energy during the year 2009.
- Aim: To increase usage of biofuels in the energy and transportation sectors of the country in the coming decade; and to utilize, develop and promote domestic feedstock and its utilization for production of biofuels.
- Biofuels Covered: Bioethanol, Biodiesel and BioCNG.
- Implementation: By National Bio-fuel Coordination Committee (NBCC), set up in 2020, chaired by Minister Petroleum and Natural Gas.
- It has representatives from 14 other ministries and departments.

- Functions of the committee:
- Take decisions for effective implementation and monitoring of biofuels program in the country, and
- Provide overall coordination among FCI and oil marketing companies.
- Policy envisages an indicative target of 20% blending of ethanol in petrol by 2030 [now amended to 2025] and 5% blending of biodiesel in diesel by 2030.
- The Policy categorizes biofuels as: -
 1. Basic Biofuels – First Generation (1G) bioethanol & biodiesel
 2. Advanced Biofuels – Second Generation (2G) ethanol, Municipal Solid Waste to drop-in fuels

- [Drop-in fuels: These are fuels derived from biomass, Agri-residues, MSW, plastic wastes, etc., that can be used in existing engines without having to modify their fuel distribution system]
 - Other advanced biofuels- bio-methanol, dimethyl ether (DME) derived from bio-methanol, bio-hydrogen
3. Third Generation (3G) biofuels, bio-CNG etc.

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- The Policy expands the scope of raw material for ethanol production by allowing use of-
- Sugarcane Juice, Sugar containing materials like Sugar Beet, Sweet Sorghum, Starch containing materials like Corn, Cassava, etc.
- Damaged food grains like wheat, broken rice, Rotten Potatoes, unfit for human consumption.
- Surplus food grains with the approval of National Biofuel Coordination Committee.
- To develop the National Biomass Repository by conducting appraisal of biomass across the country.

- Recent Amendments:
- Advance the target of 20% ethanol blending in petrol to Ethanol Supply Year (ESY) 2025-26 from earlier 2030. [ESY- 1st November of a year to 31st October of next year]
- Use of more feedstock for production of biofuels.
- Production of biofuels under the 'Make in India' programme in Special Economic Zones and Export Oriented Units.
- To grant permission for export of biofuels in specific cases.
- Addition of new members to the NBCC and it has the permission to change the policy.

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INDIA'S BIOFUEL TIMELINE



Ethanol Blending Programme introduced in 2003 by Ministry of Petroleum and Natural Gas

In 2014, Government introduced the Administered Price Mechanism



National Biofuel Policy was introduced in 2018 in line with EBP and Interest Subvention scheme introduced for enhancement of ethanol production capacity

In 2020, National Biofuel Coordination Committee to utilize surplus rice lying with FCI to be released to the distillers for ethanol production

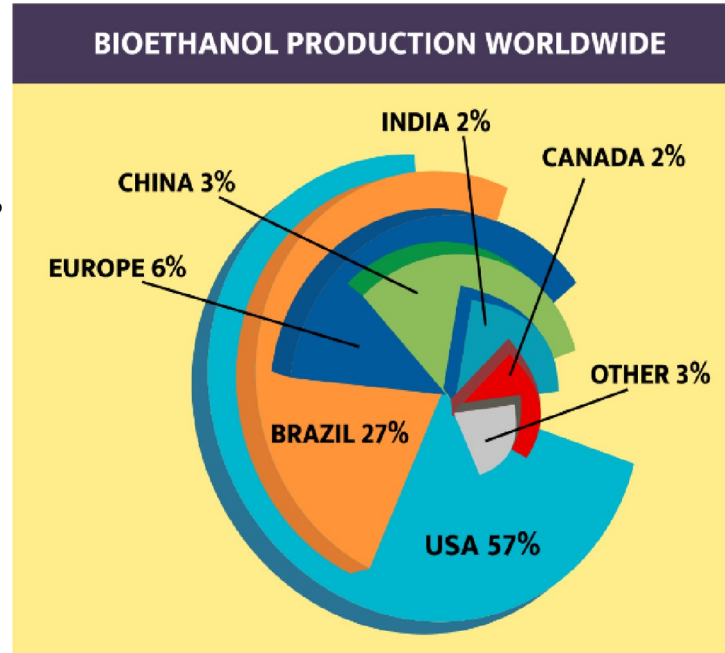


In 2021, Interest Subvention Scheme for enhancement and augmentation of ethanol production capacity was extended to grain based distilleries and distilleries producing from feedstock like sorghum, sugarbeet etc.

- **Bioethanol**
- Bioethanol (aka ethanol or ethyl alcohol (C_2H_5OH)) is an alcohol produced from starch and sugar crops.
- It is mainly produced by fermentation and by reacting ethylene with steam.
- Ethanol is a clear, colourless liquid.
- It is biodegradable, low in toxicity and causes little environmental pollution. It burns to produce carbon dioxide and water.
- It is a high-octane fuel and has replaced lead as an octane enhancer in petrol.
- By blending ethanol with gasoline, we can also oxygenate the fuel mixture, so it burns more completely and reduces polluting emissions.

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- **Produced from:**
- sugar such as sugar cane, sugar beet, sorghum, etc.;
- starch such as corn, cassava, potatoes, algae, etc.;
- cellulosic materials such as bagasse, waste wood, agricultural/forestry residues, etc. or



- **Linkages between sugar surplus and ethanol**
- As the sugarcane cultivators produce a perishable crop, they require some price guarantee. For this the central government announces **Fair and Remunerative Price** (like Minimum Support Price).
- FRP is the minimum price the **sugar mills** have to pay to the cultivators as the mill purchase the sugarcane.
- Higher cost of **raw material (sugarcane)** and cost of sugar production has affected the profits of the mills, as the mills incur losses they are not able to pay their dues to the farmers.
- Also FRP and a guaranteed procurement has increased the cultivation of sugarcane affecting **groundwater table** in many regions of India. (Ex-Ralegaon Siddhi)

- With the increasing acreage and production, sugar market has seen a **glut** bringing down the price of sugar.
- **Sugar levy**
- The **Rangrajan Panel** in 2012 had recommended complete decontrol of the sugar industry, dispensing immediately with the levy sugar obligation and administrative control on non-levy sugar.
- [Every sugar mill mandatorily surrenders 10% of its production to the central government at a price lower than the market price – this is known as levy sugar]
- **Export restrictions** (WTO norms) and price competitiveness from markets like Brazil has led to losses of sugar mills.

- In this context, **diversion of sugar surplus** to produce **bioethanol** is beneficial as
 - Solves the **liquidity** issues of sugar mills
 - Less **imports** of oil and lesser emissions of blended fuel
 - Viable sugarcane and sugar industry

- **Ethanol Blending Programme**
- An ethanol blend is defined as a **blended motor fuel containing ethyl alcohol that is at least 99% pure**, derived from agricultural products, and blended exclusively with gasoline.
- Since it is plant-based, it is considered to be a **renewable fuel**.
- Government has allowed ethanol production/ procurement from sugarcane-based raw materials viz. **C & B heavy molasses, sugarcane juice / sugar / sugar syrup, surplus rice** with Food Corporation of India (FCI) and Maize.
- The Oil Marketing Companies are to procure ethanol from domestic sources and blends ethanol at its terminals.
- Government has been notifying administered price of ethanol since **2014**.

- Department of Food and Public Distribution (DFPD) is the nodal department for promotion of fuel grade ethanol producing distilleries in the country.
- **Amended targets**
 - Recently, the central government has advanced the target of 20% ethanol blending in petrol (also called as **E20**), by five years to 2025, from 2030.
 - The Government has also released an expert committee report on the **‘Roadmap for Ethanol Blending in India by 2025’**.
 - The committee recommended introducing vehicles that are compatible by rolling out of E20 material compliant and E10 engine-tuned vehicles from April 2023 and production of E20-tuned engine vehicles from April 2025.

- **Recent Initiatives to promote bioethanol**
- Under **PM-JIVAN (Jaiv Indhan- Vatavaran Anukool fasal awashesh Nivaran) Yojana**, 12 commercial plants and 10 demonstration plants of Second Generation (2G) Bio-Refineries are envisaged to be set up in areas having sufficient availability of biomass so that ethanol is available for blending throughout the country.
- Cabinet Committee on Economic Affairs (CCEA) approved ₹8,460 crore **Modified scheme for extending interest subvention** for those setting up standalone ethanol distilleries using grain, molasses, dual feed, sugar beet, sweet sorghum and cereals as a feedstock.
- The focus is on increasing India's ethanol production capacity.
- Prime Minister has launched a **Pilot Project of E100** dispensing stations at three locations in Pune.

- **Recent Initiatives to promote bioethanol**
- **Notification of Mass Emission Standards for E12 AND E15 Fuels** by Ministry of Road Transport and Highways
 - The mass emission standards for E 12 (12% Ethanol with Gasoline) and E15 (15% Ethanol with gasoline) fuels are notified under the Central Motor Vehicles (Twenty Fifth Amendment) Rules, 2021.
 - This will enable the Automotive Industry to manufacture E 12 and E 15 compliant motor vehicles.

- **Cassava (Tapioca)**
 - **ICAR-Central Tuber Crops Research Institute (CTCRI)** has identified Cassava (tapioca) as a promising raw material for bioethanol production to meet **Ethanol Blending Petrol (EBP) programme target of 2025.**
- **About Cassava:**
 - Its starch with its unique physico-chemical and functional properties finds extensive applications in the food and industrial sectors.
 - The agricultural residues of cassava such as peels, stems and leaves are potential feedstock for **2G bioethanol production.**
 - Major production is from Tamil Nadu, followed by Kerala.

Flexi Fuel

- **Context:** Ministry of Road Transport & Highways has launched first of its kind pilot project on Flexi-Fuel Strong Hybrid Electric Vehicles (FFV-SHEV) in India.

Flex Fuel vehicles (FFV)

- Like traditional vehicles, flex fuel vehicles have an internal combustion engine, but instead of regular petrol, it can run on **blended fuel — petrol with ethanol or methanol.**
 - The ethanol mix can vary between 20% and 85%.
- Unlike electric hybrid vehicles, no bulky parts need to be added to the basic gasoline vehicle architecture.
- Upgrading existing vehicles is possible but expensive and not considered feasible.

- Flexi Fuel Vehicles are currently being **used successfully in Brazil**, giving people the option to switch fuel (gasoline and ethanol).
- An **Flexi-Fuel Strong Hybrid Electric Vehicles** (FFV-SHEV) possesses a flex-fuel engine and an electric powertrain.
 - This setup extends dual advantages of higher ethanol use and greater fuel efficiency, as it can run on its EV mode for a good amount of time, while the engine stands shut off.
- **Significance of FFV:** Less polluting, check on diversion of sugar used in ethanol blending, burn with whatever proportion of mixture is in its combustion chamber, reduced import bill etc.
- **Challenges of FFV:** lack of adequate infrastructure, issue with ethanol supply since this largely comes from sugarcane in India, **less mileage** etc.

Different types of vehicles as per fuel



Mono-fuel vehicle

Vehicle which operates on natural gas only. Also known as Dedicated Natural Gas Vehicle.



Bi-fuel vehicle

Vehicle that has two independent fuel systems (one of them for natural gas) and can run on both fuels simultaneously.



Flex fuel vehicles (FFVs)

FFV is any motor vehicle (or motor vehicle engine) engineered and designed to be operated on any mixture of two or more different fuels.

- FFVs are capable of running on any blend of gasoline and ethanol up to 83%.

Fuel Cell

- Electrochemical devices to convert chemical energy into electrical energy.
- Fuel cells **do not need to be periodically recharged** like batteries, but instead continue to produce electricity as long as it is supplied with a source of hydrogen and a source of oxygen (usually air).
- Source of **hydrogen is generally referred to as the fuel** and this gives the fuel cell its name.
- **Applications:** Transport, Portable (military applications, Auxiliary Power Units, laptops etc.), Stationary (power to industrial/commercial/residential buildings etc.),

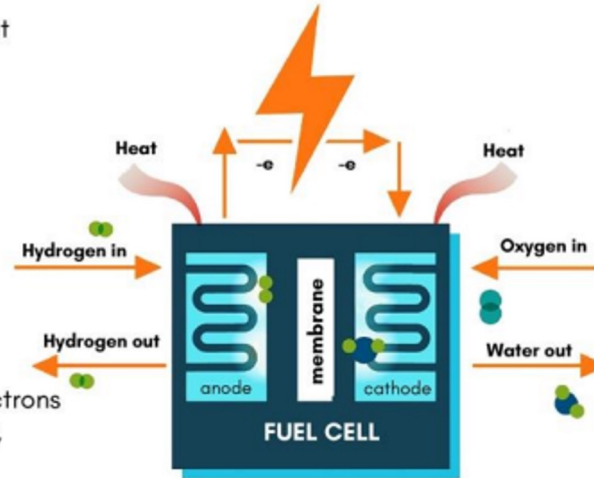
Renewable Energy



- **Benefits:** Lower operational costs due to higher efficiency and high energy density of hydrogen, Noise free operation, operating times are much longer.
- **Challenges:** Transportation, flammability and storage of hydrogen gas, High cost, Technological challenges like system size, management of air, heat and water etc.

How Fuel Cells Work

- 1 The hydrogen atoms enter at the **anode**.
- 2 The atoms are stripped of their electrons.
- 3 The positively charged protons pass through the membrane to the **cathode**.
- 4 The negatively charged electrons are forced through a circuit, generating electricity.
- 5 After passing through the circuit, the electrons combine with protons to generate the fuel cell's byproducts: **water** and **heat**.



Roshini

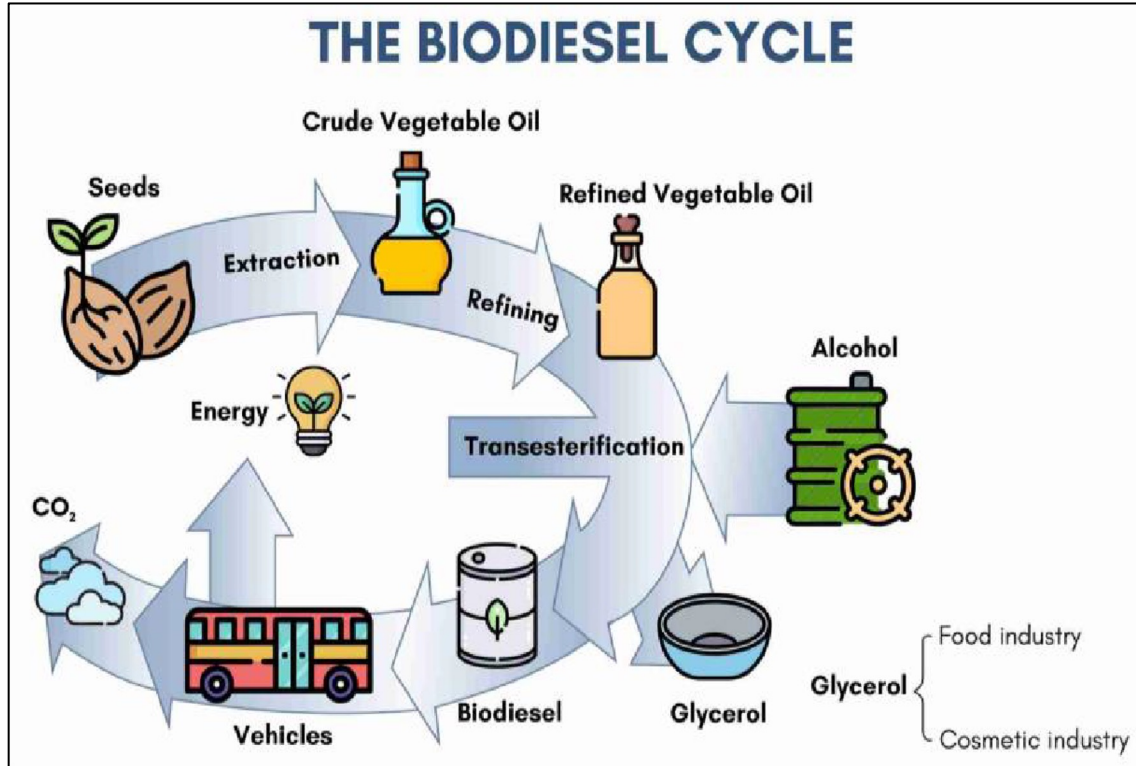
- It is India's **first Saline Water Lantern** which uses sea water as electrolyte between specially designed electrodes to power the LED lamps.
- Developed by: **National Institute of Ocean Technology (NIOT), Chennai.**
- Technology can also be used in hinterlands, as any saline/normal water mixed with common salt can be used to power the lantern.

Biobutanol

- Biobutanol is **four-carbon alcohol** produced by the fermentation of biomass.
- The production of biobutanol can be carried out in ethanol production facilities.
- Its properties are **similar to that of gasoline**.
- Some gasoline-powered vehicles can even use biobutanol without being modified.
- However, it has a lower energy content, on average 10-20%, than that of gasoline, which is a major **disadvantage** of biobutanol.
- Biobutanol exhibits the potential to reduce carbon emissions by **85% when compared to gasoline**, thus making it a viable and suitable alternative to gasoline and gasoline-ethanol blended fuels.

- **Biodiesel**
- Biodiesel is made from renewable sources such as **vegetable/plant/animal oils for use in diesel engines.**
- Vegetable oils are chemically called **triglycerides** (fats).
- Biodiesel comprises esters of **long chain fatty acids** derived from these oils.
- To make biodiesel, fats in the vegetable oil (triglycerides) are reacted with alcohol — usually **methanol**.
- In this reaction, glycerine (in triglycerides) is replaced by methanol to produce **methyl ester (biodiesel)**.
- Biofuel development in India centres around the cultivation of Jatropha plant seeds — rich in oil (40%).

Renewable Energy



- **Biodiesel blend**
- The biodiesel blend is a mix of biodiesel with fossil fuel, designated as BXX, where XX represents the volume percentage of biodiesel in the blend (B100 means pure biofuel).
- Currently, biodiesel is blended with **the fossil fuel in the proportion of 2%, 5%, etc**, although technically it can be used as a pure fuel with some minor modifications in existing engine systems.
- **Advantages of biodiesel**
- Biodiesel has intrinsic lubricating properties (diesel engines are long lasting compared to petrol engines because of this very property)

- The **Cetane Index (CI)**, a measure of the inflammability of fuel, is more than 56 to 58 for biodiesel compared 50/52 for fossil fuel. A higher CI value will mean better ignition and combustion.
- The biodiesel molecule contains about **11% oxygen**, facilitating improved combustion and less soot.
- The sulphur content in biodiesel is as low as 0.001%.
- Biodiesel requires less energy to produce than fossil fuels (for every unit of energy needed to produce biodiesel, 3.24 units of energy are gained)
- Biodiesel production does not hurt the production of edible oil ('food or fuel' dilemma doesn't exist).

- **Used Cooking Oil (UCO) based Biodiesel**
- **Context-** Recently, Indian Oil Corporation has kicked-off the supply of diesel that is blended with biodiesel made from used cooking oil.
- **About Used Cooking Oil (UCO)**
- UCOs are oils and fats **that have been used** for cooking or frying in the food processing industry, restaurants, fast foods and at consumer level, in households.
- UCO must contain **only fats, oils, or greases** that were previously used for cooking or frying operations.
- UCO is an important source of raw material to produce biodiesel.
- It is also used for making soap, cosmetics, cooking oil, and animal feed, etc.

- **Benefits of biodiesel made from UCO**
- Prevent people from serious ailments such as hypertension, obesity, atherosclerosis, Cancer, Alzheimer's disease, liver diseases by removing reused or burnt cooking medium from the food chain.
- UCO when discarded without any treatment **clogs drainage systems.**
- Contributing to the needs of a **Circular Economy.**
- **Initiative taken to make biodiesel from UCO**
- In 2019, Ministries of Petroleum and Natural Gas & Steel, along with Health & Family Welfare, Science & Technology and Earth Sciences, had initiated Expressions of Interest for "Procurement of Bio-diesel produced from UCO"

- It aims to create an ecosystem for collection and conversion of UCO into Biodiesel and developing entrepreneurship opportunities.
- Under this initiative, **OMCs offer periodically incremental price guarantees for five years** and extend off-take guarantees for ten years to prospective entrepreneurs.
- **National Policy on Biofuels, 2018**, encourages setting up of supply chain mechanisms for biodiesel production from non-edible oilseeds, Used Cooking Oil, short gestation crops.
- **Food Safety and Standards Authority of India (FSSAI)** in association with the **Biodiesel Association of India (BDAI)** launched **‘Repurpose Used Cooking Oil (RUCO) project’** in 2019.

Renewable Energy



- Project is aimed at purchasing used oils from hoteliers, caterers, snack makers and traders at a reasonable price and converting it into biodiesel at a plant.
- RUCO sticker and a mobile phone application was launched for collection of used cooking oil (UCO) to ensure that it does not come back to ecosystem.

- **Bio-Jet Fuel Technology**
- **Context- CSIR-IIP Dehradun's** home-grown technology to produce bio-jet fuel, formally approved for **use on military aircraft of the Indian Air Force.**
- Bio-jet fuel can be produced from used cooking oil, tree-borne oils, short gestation oilseed crops, and waste extracts from edible oil processing units.
- Earlier, AN 32 (transportation plan) and commercial plan (Spice jet) was flown in 2018 using the bio-jet fuel.

Compressed BioGas (CBG)

- **Context:** Union Minister of Petroleum and Natural Gas inaugurated Asia's largest CBG plant in Sangrur, Punjab.
- CBG is the compressed and purified bio-gas produced from waste/bio-mass sources like agricultural residue, cattle dung etc.
- **Benefits:**
 - **Reduce the problem of Stubble Burning and related carbon dioxide emissions.**
 - Produce fermented organic manure for organic farming.
 - Generate employment and additional income for farmers.

Biomass co-firing

- **Context:** Government had launched the **SAMARTH Mission** i.e., National Mission on Use of Biomass in coal based thermal Power Plants.
- **Co-firing biomass** consists of burning biomass in coal-fired power plants along with coal. Three different concepts for co-firing i.e, Direct co-firing, Indirect co-firing, Parallel co-firing.
- **Advantages:** **lower** capital costs, higher efficiency, increased economies of scale, **Lower electricity costs** due to the larger size and superior performance of conventional coal power plants

- **National Bioenergy Programme (NBP)**
- Context
- Recently, Ministry of New and Renewable Energy (MNRE) has notified that National Bioenergy Programme (NBP) will continue from FY 2021-22 to 2025-26 with a budget outlay of ₹858 crore for the first phase.
- NBP will comprise the three sub-schemes
 1. Waste to Energy Programme
 2. Biomass Programme
 3. Biogas Programme.

Renewable Energy

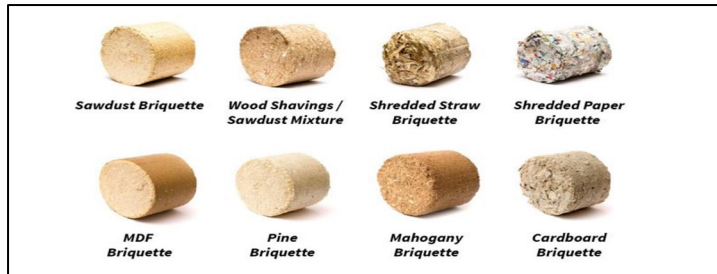


- **Terms**
- **Biomass** is renewable organic material that comes from plants and animals. Biomass is the raw material to produce biogas. Major component of biomass is **carbon**. [Recall carbon cycle]
- **Bioenergy** refers to electricity and gas that is generated from organic matter, known as biomass.
- **Biogas** is a renewable fuel produced by the breakdown of organic matter such as food scraps and animal waste by microorganisms in absence of oxygen.
- Biogas is composed mostly of **methane (CH₄)**, and **carbon dioxide (CO₂)**.

- **Sub-scheme 1: Waste to Energy Programme (Programme on Energy from Urban, Industrial and Agricultural Wastes /Residues)**
- Objective: To support the setting up of **Waste to Energy** projects for generation of Biogas/ Bio-CNG/ Power plants (excluding municipal solid waste to power project)/ producer or syngas from urban, industrial and agricultural wastes/residues.
- **Implementing agency:** Indian Renewable Energy Development Agency (IREDA).
- IREDA, established in 1987, is a **Mini Ratna** (Category-I) Enterprise under administrative control of **MNRE**.

- **Scope**
- Provides Central Financial Assistance (CFA) to project developers and service charges to implementing/inspection agencies in respect of successful commissioning of waste to energy plants.
- **Advantages**
- Safe, technologically advanced means of waste disposal that reduces greenhouse gases, generates clean energy and recycles metal.

- **Sub-scheme 2: Biomass Programme**
- Also called Scheme to Support Manufacturing of Briquettes & Pellets and Promotion of Biomass (nonbagasse) based cogeneration in Industries
- Objective
- To support setting up of Biomass Briquette/Pellet manufacturing plants and to support Biomass (non-bagasse) based cogeneration projects in Industries.
- [A briquette is a compressed block of coal dust or other combustible biomass material]
- Implementing agency: IREDA.



- **Scope**
- Provides Central Financial Assistance (CFA) to project developers and service charges to implementing agency and inspection agencies in respect of setting up of Briquette / Pellet manufacturing plants and Biomass (nonbagasse) cogeneration projects in industries.
- **Advantages**
- Reduce stubble burning. [More discussion to be followed while discussing stubble burning]
- Provide additional source of income to farmers and better environmental practices and reduced pollution.

- **Sub-scheme 3: Biogas Programme**
- Objective
- Setting up of biogas plants for small power needs of users.
- Implementation
- By designated Programme Implementing Agency (PIA) of State/ Union Territory/Biogas Development and Training Centres (BDTC).
- IREDA, NABARD/ RBI approved Financial Institutions may also implement it in consultation with PIAs.

- **Scope**
- Covers setting up of small and medium biogas plants ranging from 1M3 (cubic meter) to 2500 M3 biogas generation per day for individual user, farmers, poultry etc.
- **Advantages**
- GHG reduction, improved sanitation, Employment Creation, and digested slurry from biogas plants can supplement / reduce use of chemical fertilizers.

National Green Hydrogen Mission

why is
 H_2 important

types of
 H_2

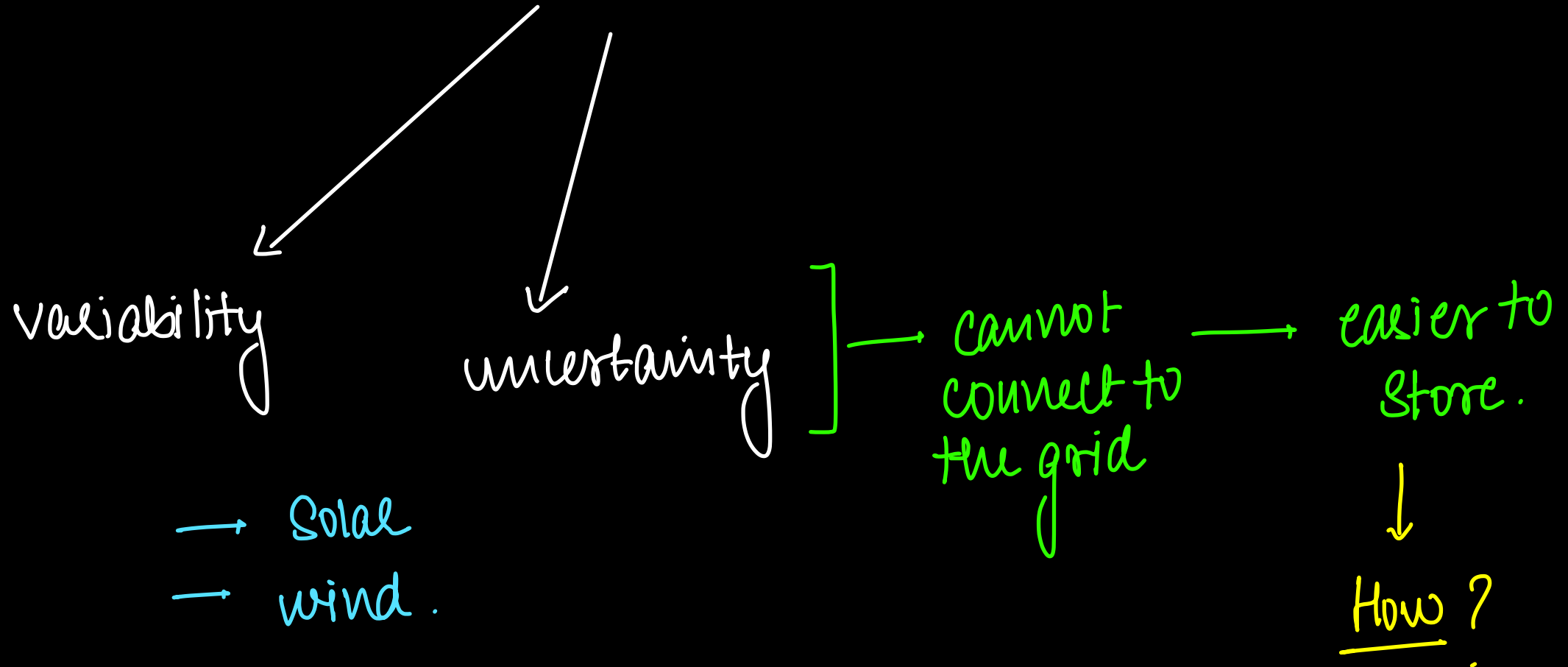
CCUS

advantages and
disadvantages

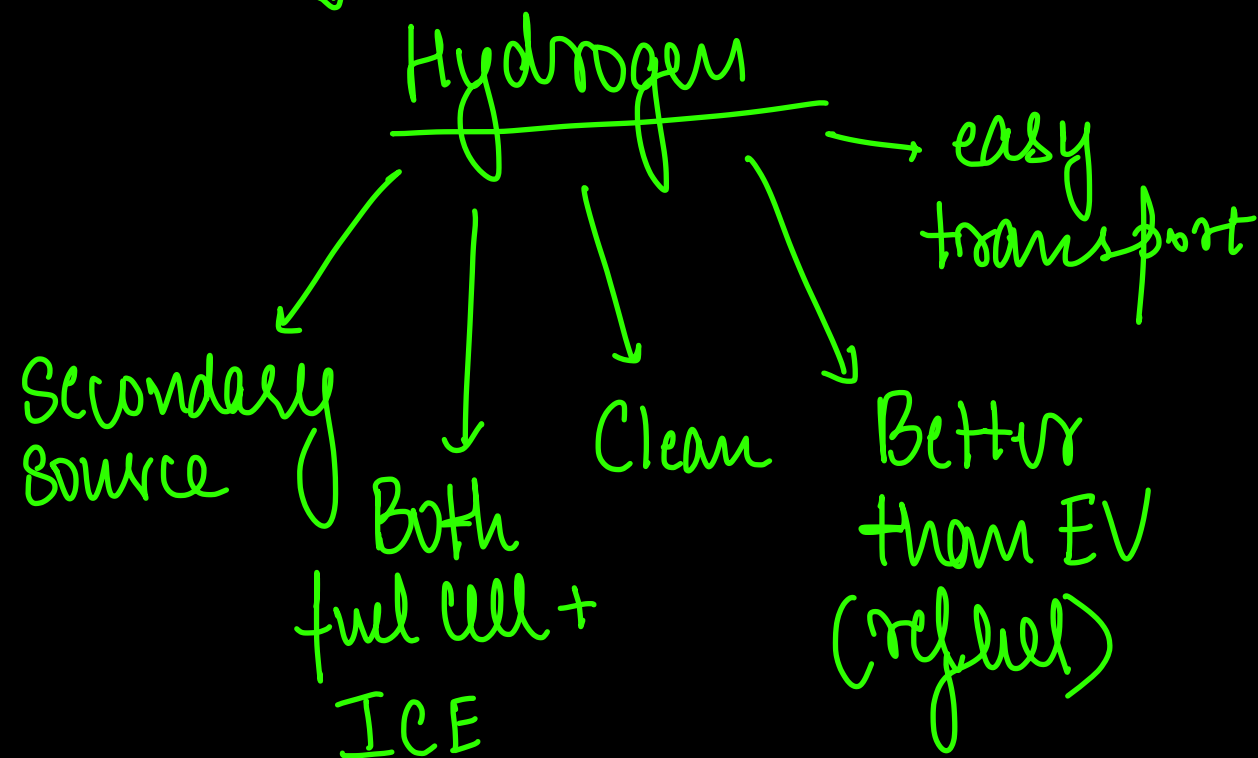
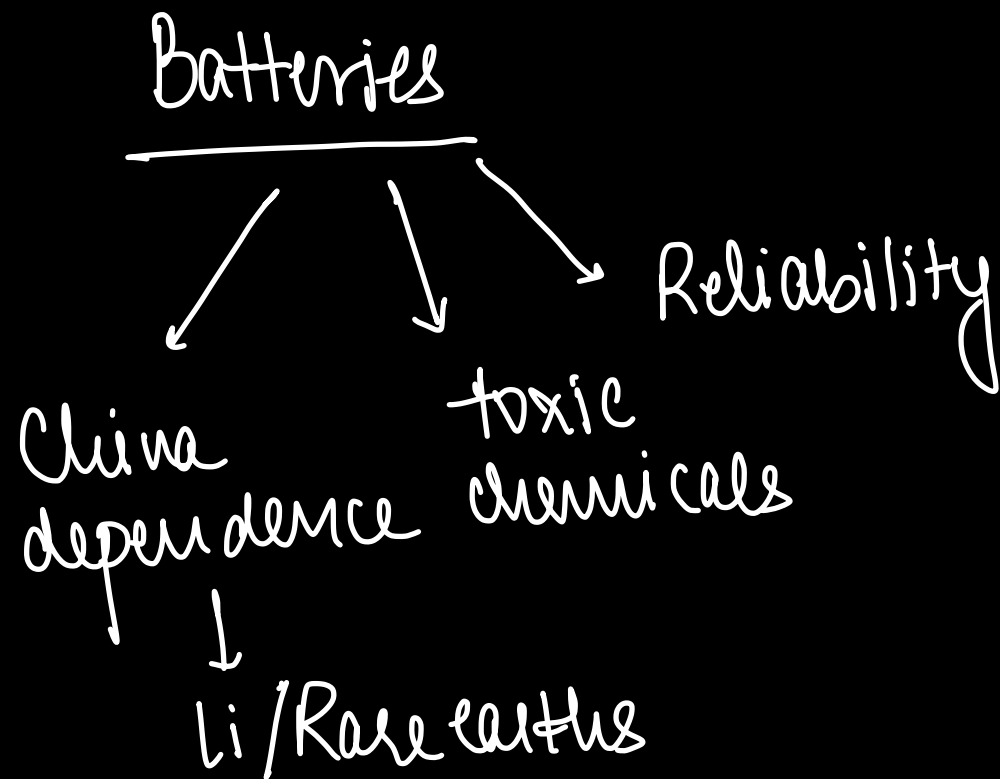
India's commitments under the UNFCCC : Panchamrita

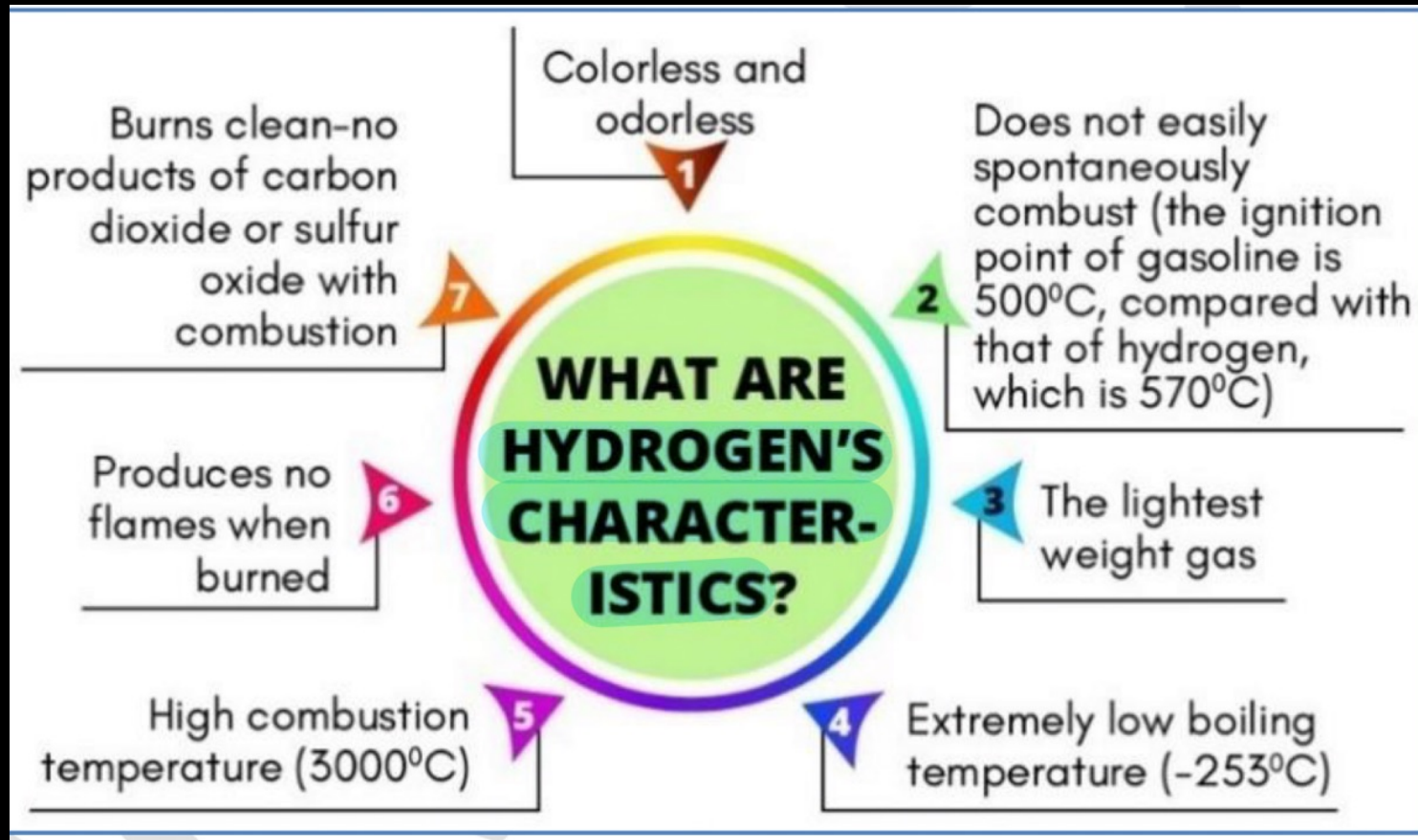
- i. Reach 500GW Non-fossil energy capacity by 2030. Renewable energy
 - ii. 50 per cent of its energy requirements from renewable energy by 2030. RE
 - iii. Reduction of total projected carbon emissions by one billion tonnes from now to 2030. RE
 - iv. Reduction of the carbon intensity of the economy by 45 per cent by 2030, over 2005 levels.
 - v. Achieving the target of net zero emissions by 2070. RE
- Renewable energy

Issues with Renewables



Storing Renewables

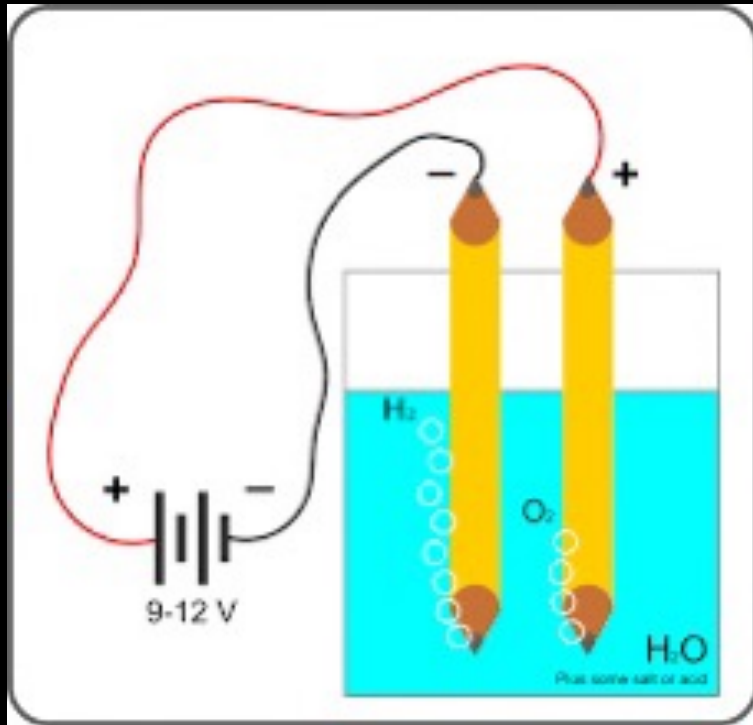
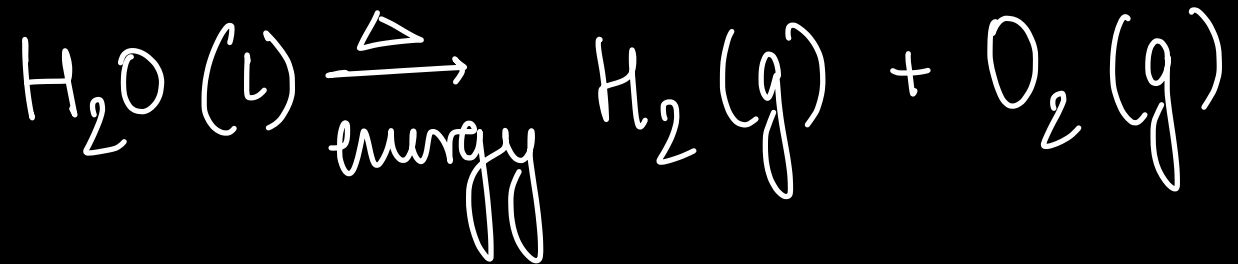




How to
Make
Hydrogen?
↓
electrolyser



Making Hydrogen



Energy → from various sources
→ store it in Hydrogen.
→ depending on type of energy, different colors.

Types of Hydrogen

GREEN

Hydrogen produced by electrolysis of water, using electricity from renewable sources like hydropower, wind, and solar. Zero carbon emissions are produced.

TURQUOISE

Hydrogen produced by the thermal splitting of methane (methane pyrolysis). Instead of CO₂, solid carbon is produced.

YELLOW

Hydrogen produced by electrolysis using grid electricity.

BLUE

Grey or brown hydrogen with its CO₂ sequestered or repurposed.

PINK/PURPLE/RED

Hydrogen produced by electrolysis using nuclear power.

BLACK/GRAY

Hydrogen extracted from natural gas using steam-methane reforming.

WHITE

Hydrogen produced as a byproduct of industrial processes.

BROWN

Hydrogen extracted from fossil fuels, usually coal using gasification.

Green Hydrogen

Simplest

use Renewable energy for electrolysis of water

Other types of H₂ ?

pink

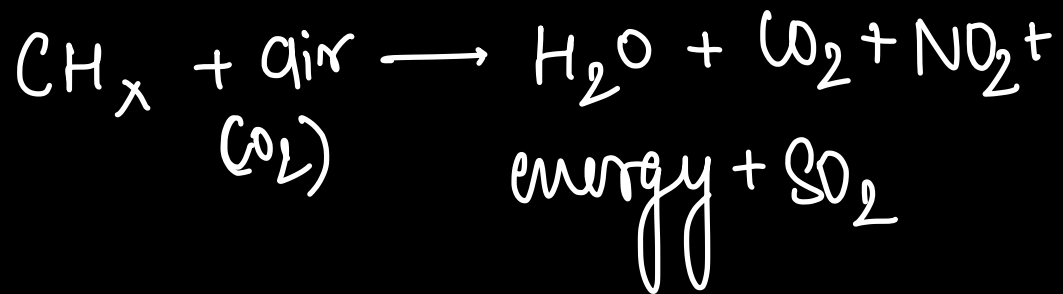
yellow

white.

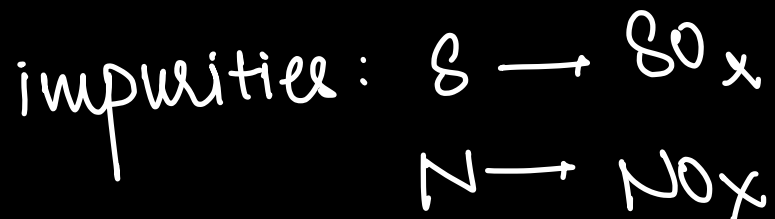
For others, we need to understand further

Methane / coal Based H_2

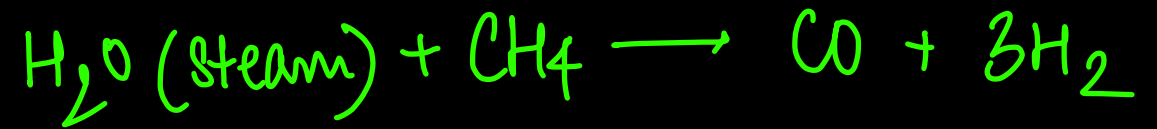
Normal process



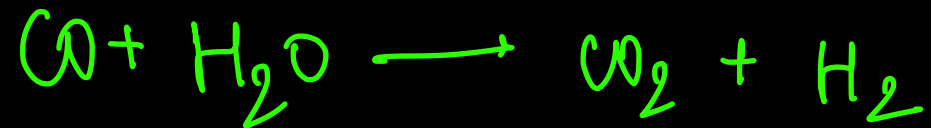
issue:



Hydrogen production from Natural gas



Steam Methane Reforming



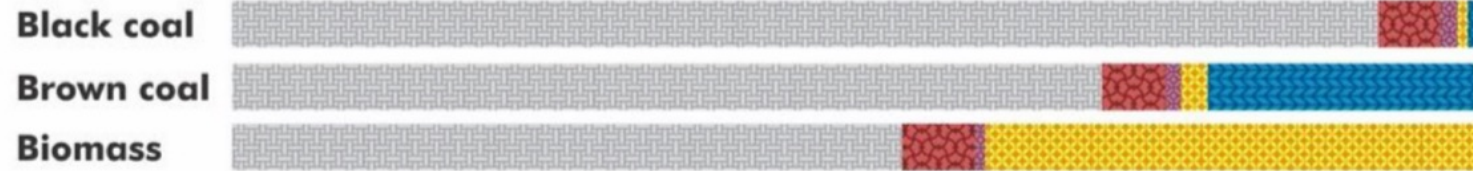
water gas shift RK^n

GRAY H_2

Brown Hydrogen: coal

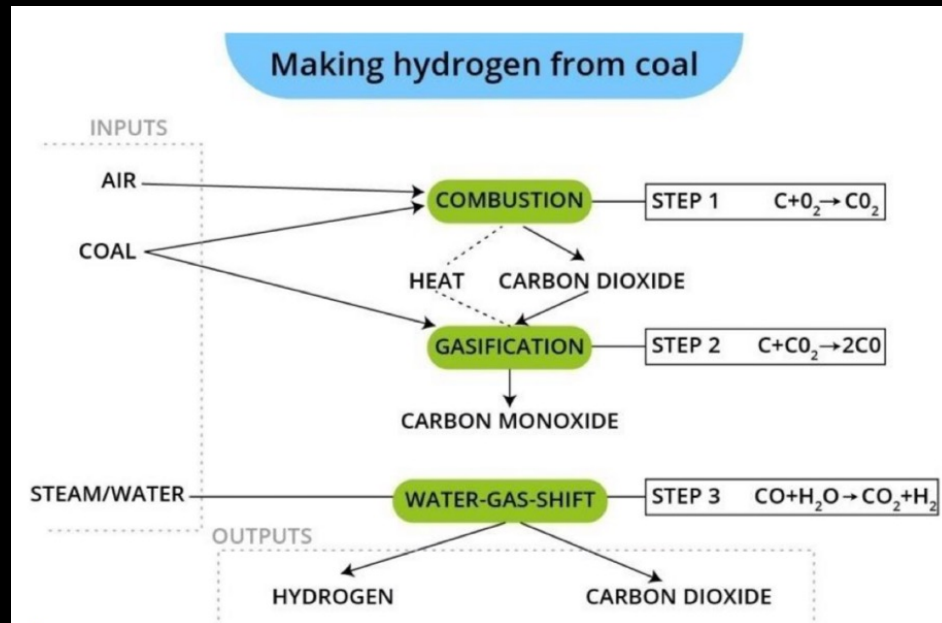
Coal Composition

Carbon Hydrogen Nitrogen Sulfur Oxygen



coal has carbon + impurities

Normal coal Burn → pollution

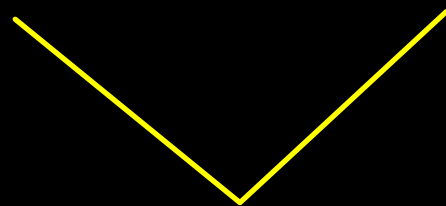


coal gasification
+
water gas shift

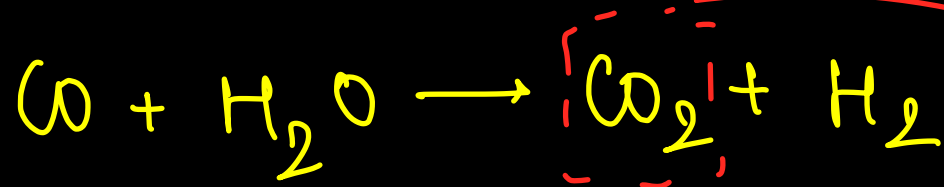
→ Brown H_2

Brown H_2

Gray H_2



final step:



pollutant

also in other combustion
 $2K^n$

greenhouse gas

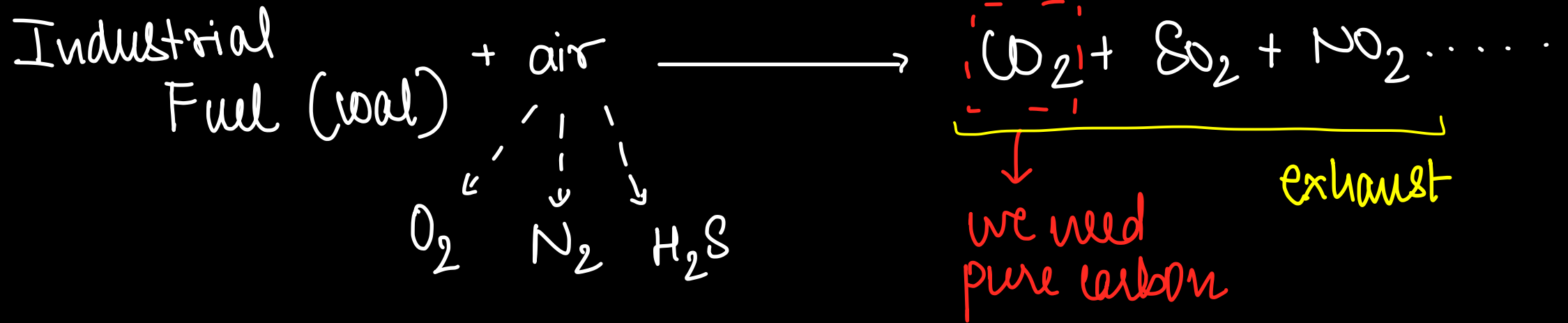
combustion of coal $\rightarrow$ impurities also $\rightarrow CO_2 + SO_x + NO_x$

Carbon capture utilization and Storage (CCUS)

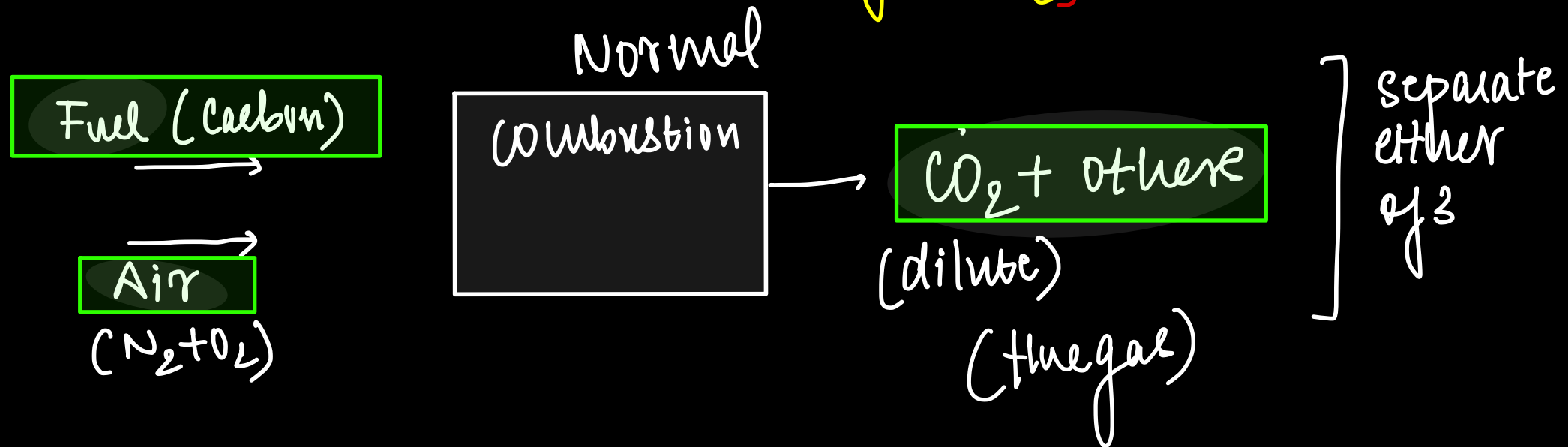
capturing carbon
from coal plants/
combustion

reducing
carbon
in air

Utilising the captured
carbon or
storing it



gas ~~is~~

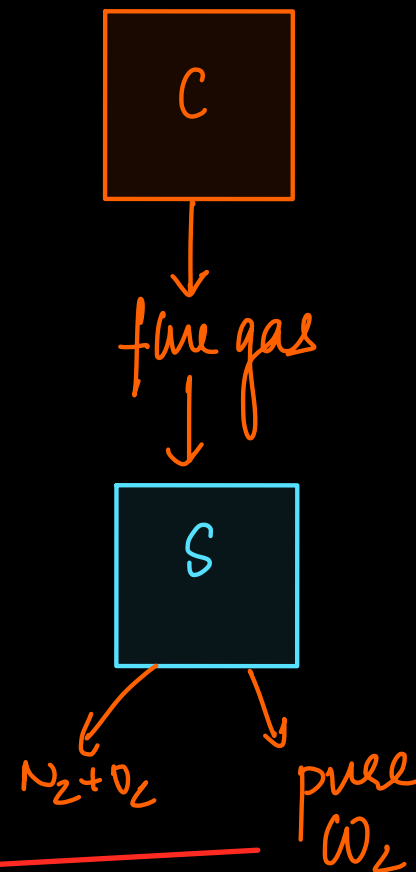
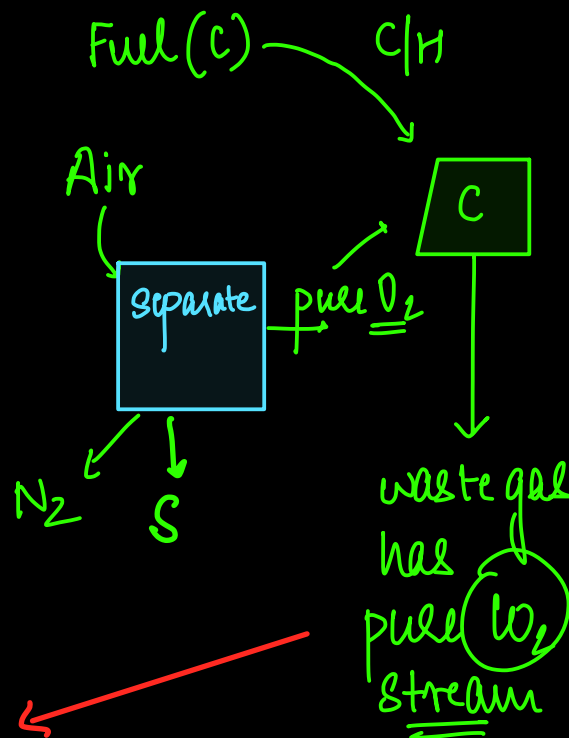
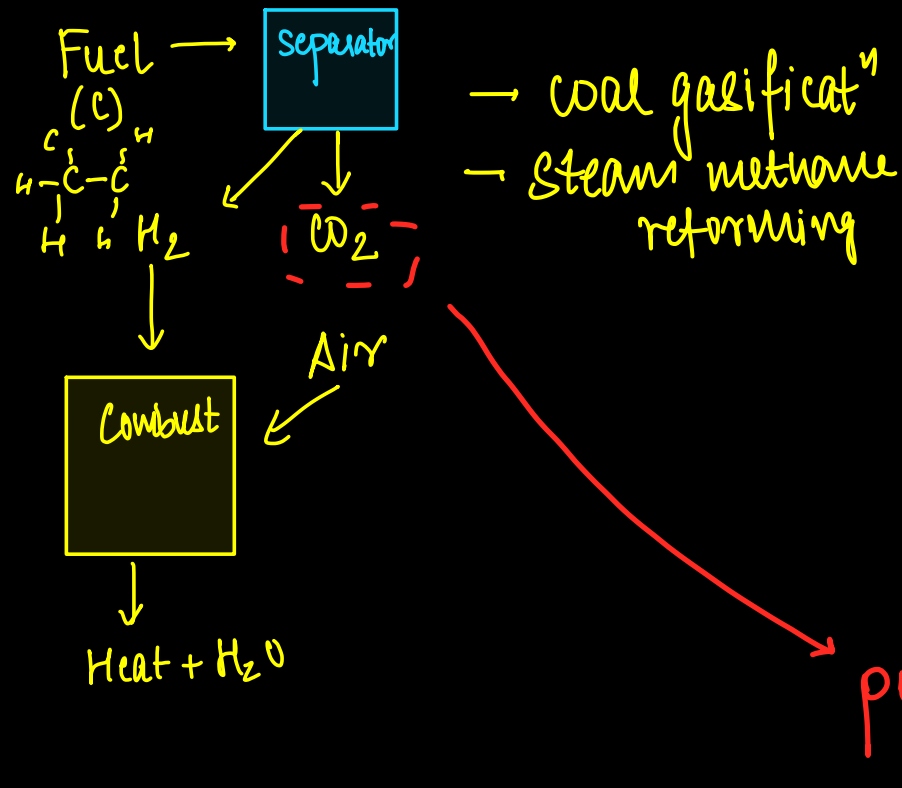


CCUS

Pre combustion capture

Oxy fuel combustion

Post combustion Retrofit



Carbon capture, utilisation and storage (CCUS)

CAPTURE

Capturing CO₂ from e.g. biomass-fuelled power stations, industrial facilities or directly from air.



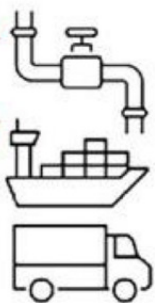
USE

Capturing CO₂ is used as a resource or feedstock to create products and services.



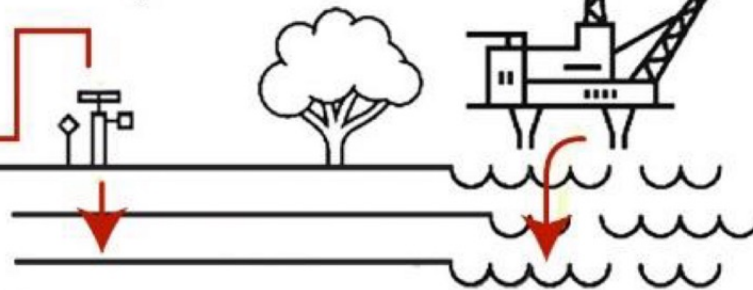
TRANSPORT

The compressed CO₂ is moved by ship, truck or pipeline from point of capture to the point of use or storage.



STORAGE

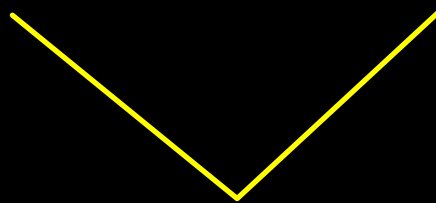
Storing the CO₂ permanently in underground geological formations.



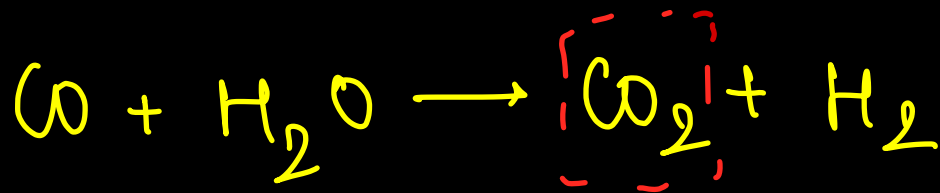
dry ice
carbonic
beverages.

↳ abandoned mines, under the sea, old reservoirs.

Brown H₂ Gray H₂



final step:



↓
CCUS
↓
Blue H₂

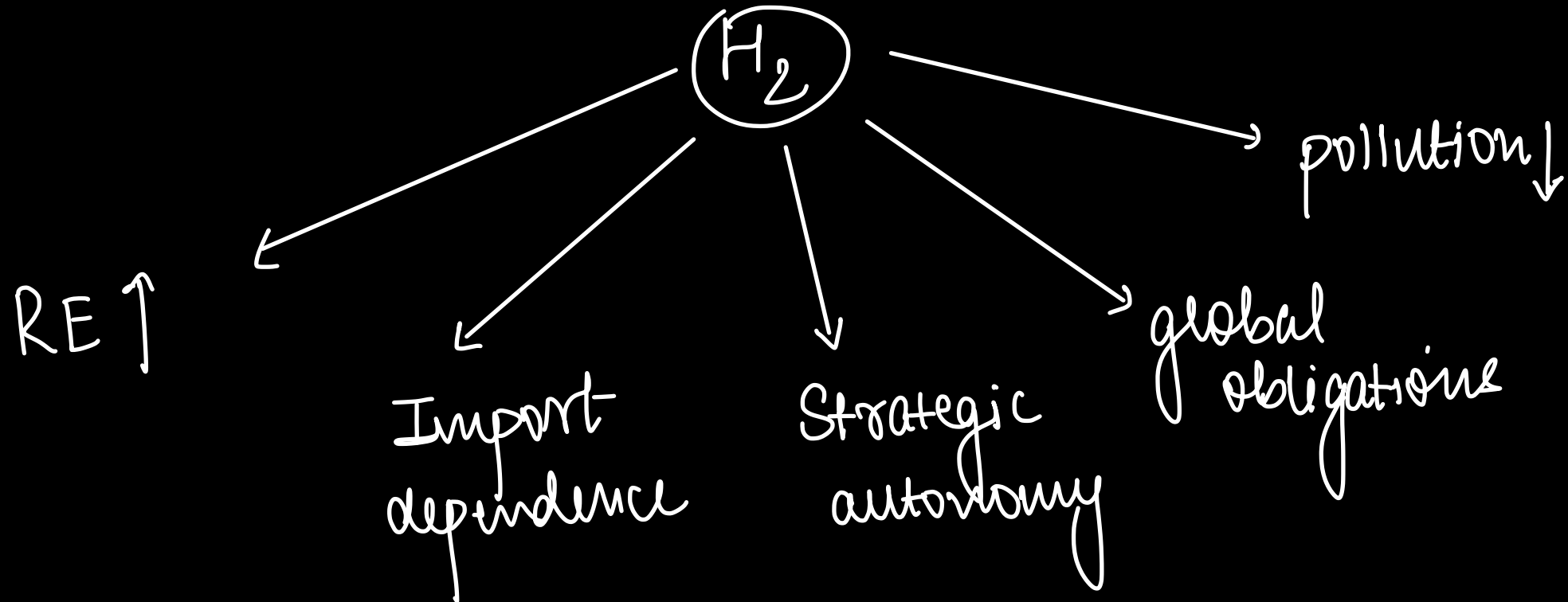


H₂ CCUS

↓
Carbon emissions

- i. Reach 500GW Non-fossil energy capacity by 2030.
- ii. 50 per cent of its energy requirements from renewable energy by 2030.
- iii. Reduction of total projected carbon emissions by one billion tonnes from now to 2030.
- iv. Reduction of the carbon intensity of the economy by 45 per cent by 2030, over 2005 levels.
- v. Achieving the target of net zero emissions by 2070.

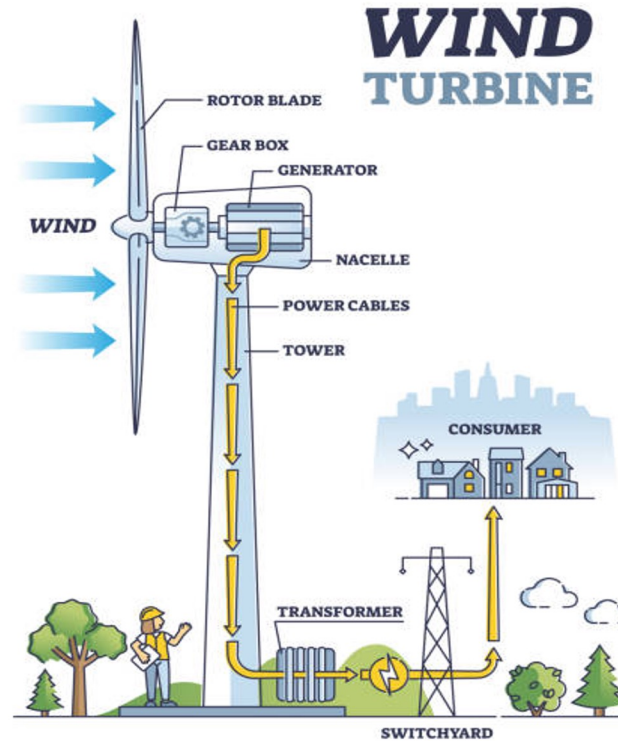
Main perspective



Renewable Energy

Wind Energy

- **Principle:** Kinetic energy of wind to mechanical energy of rotor blades which can move turbine-generator to produce electrical energy.
- It generates **AC power**
- **Gear Box:** Converts low Revolutions per minute (RPM) to high RPM.



Suitable locations for Wind Energy Plants

1. Coastal

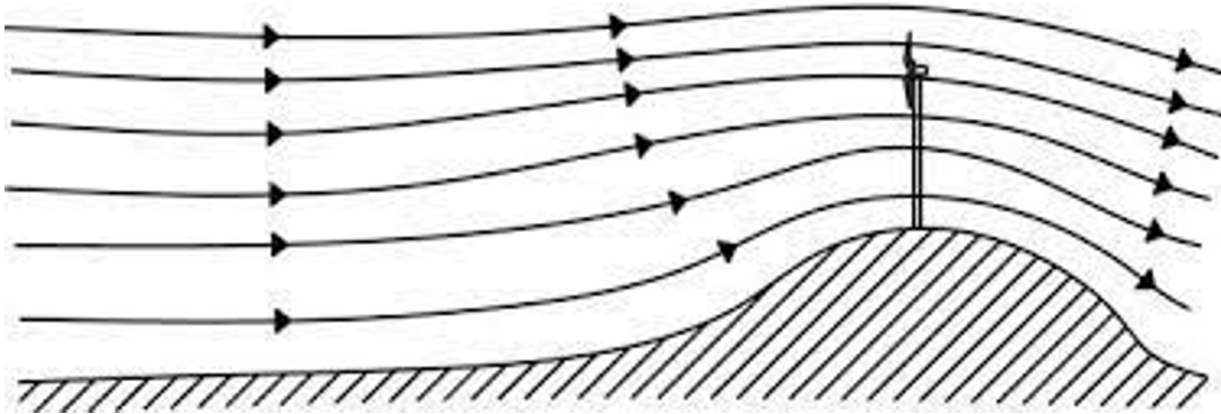
- **Onshore:** Most of the India's wind power is on shore.
- **Offshore:** They generate electricity from wind blowing across the sea. They are more efficient than onshore wind farms, due to the **higher speed of winds, greater consistency and lack of physical interference.**
 - **Pilot Projects:** Kanyakumari, Rameshwaram, etc.



Suitable locations for Wind Energy Plants

2. Hilly Region

- Favorable sites include the **tops of smooth, rounded hills and mountain gaps that funnel and intensify wind.**



Suitable locations for Wind Energy Plants

3. Islands

- Favorable sites include the **tops of smooth, rounded hills and mountain gaps that funnel and intensify wind.**
- **Andaman Nicobar Islands (longer islands) > Lakshadweep (atolls)**

4. Deserts:

- **Example: Saudi Arabia** has a potential of **200 GW** due to its Desert landscape.

Wind Aero generator

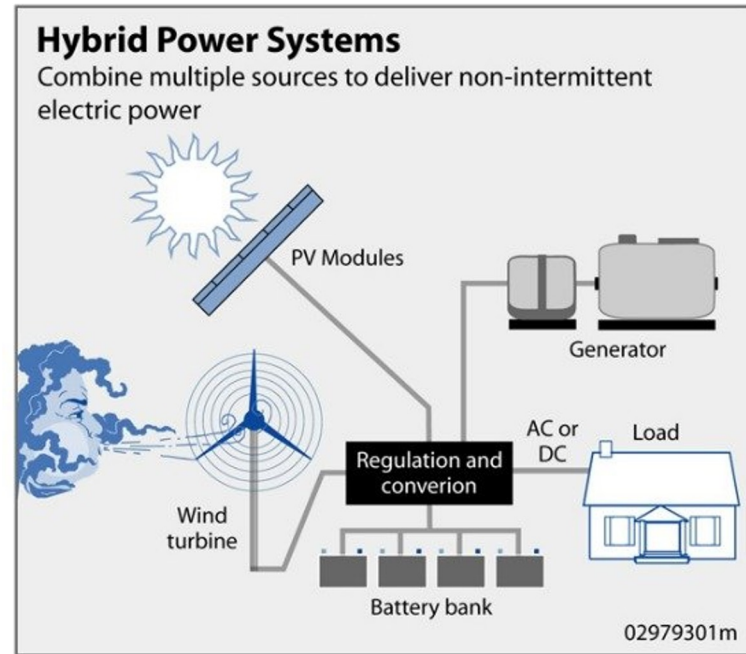
- An aerogenerator is a **small wind electric generator** having a capacity of up to **30 kW**.
- Aerogenerators are installed either in stand-alone mode (off grid) or along with solar photovoltaic (SPV) systems to form a wind-solar hybrid system for **decentralized power generation**.
- An aerogenerator is suitable for power generation in unelectrified areas having adequate wind speeds.

Renewable Energy

Wind-Solar Hybrid System

- A wind solar hybrid project is a renewable energy system that combines both wind turbines and solar panels in the same location to generate electricity. This approach is a solution to the **intermittency challenges** of both solar and wind energy generation providing a more stable and reliable energy output.
- The **MoNRE** has issued the **National Wind-Solar Hybrid Policy** in **2018** with an aim to boost renewable power generation by promoting new projects as well as hybridisation of existing ones.

Environment & Ecology by Dr. Shivin Chaudhary



[Click Here to Learn More](#)

- **Wind Energy**
- India currently has an installed capacity of 41 GW (Feb 2023), the fourth highest in the world. (**China > USA > Germany > India**)
- However, this is just **29%** of its target, which is to install **140 GW by 2030**.
- Tamil Nadu accounts for **25%** of this installed capacity followed by Gujarat at **24%**.
- **Important wind farms**
 - Muppandal Wind Farm, Kanyakumari district- 1.5 GW
 - Jaisalmer Wind Park, Rajasthan- 1.06 GW

Muppandal Wind Farm, India

- The area of Muppandal features vast barren land, which is not fit for cultivation but **boasts of high pressure winds (due to funnelling effect)** making it an ideal location for the development of wind farms.
- Muppandal receives strong winds for nine months a year from the west, due to the presence of the **Western Ghats mountain range**.

National Offshore Wind Energy Policy

- It has been notified in **2015** to provide a legal framework for development of the offshore wind sector in India.
- Policy authorises, the **Ministry of New and Renewable Energy** to work in close coordination with other government entities for development of offshore wind energy within the **entire exclusive economic zone (EEZ)** of the country.
- **National Institute of Wind Energy (NIWE)** has been designated as the nodal agency to carry out or facilitate necessary resource assessment activity for offshore wind development.

Global Offshore Wind Alliance

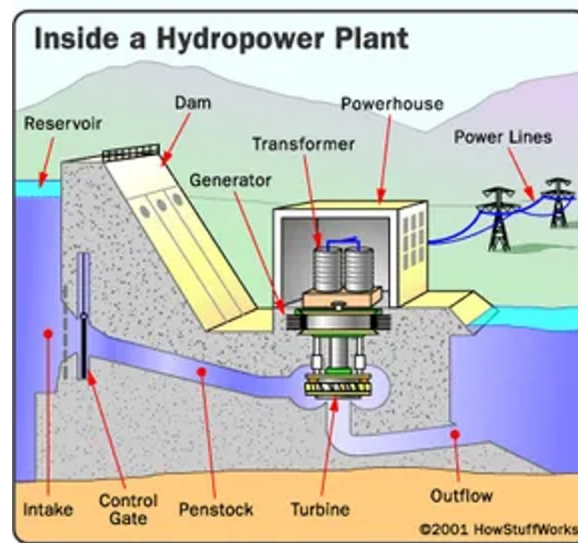
- Nine new countries including Belgium, Colombia, Germany, US, UK etc. join GOWA.
- Initiated by: **International Renewable Energy Agency (IRENA), Denmark and Global Wind Energy Council (GWEC).**
 - GWEC is international trade association for the wind power industry (India is also a member).
- Brings together governments and other stakeholders to accelerate deployment of offshore wind power.
- **Objective:** To contribute to achieving a total offshore wind capacity of a minimum of 380 GW by 2030 and an installed capacity increase of at least 70 GW per year from 2030.

Small Hydro Power

- In India, hydro power plants with **capacity of 25 MW or below** are classified as small hydro.
- Hydro Power was looked after by **Ministry of Power** prior to **1989**.
- In **1989**, plant capacity up to 3MW and below was transferred to the Ministry of New and Renewable Energy (MNRE). And, Subsequently plant capacity up to **25MW** and below was entrusted with the MNRE in November **1999**.
- **Estimated potential:** 21133 MW from 7133 sites

Renewable Energy

- The hydropower facilities are **powered by the kinetic energy** of flowing water as it moves downstream.
- Hydropower **utilizes turbines and generators** to convert that **kinetic energy into electricity**, which is then fed into the electrical grid to power homes, businesses, and industries.
- At the plant level, water flows through a **pipe—also known as a penstock**—and then spins the blades in a turbine, which, in turn, spins a generator that ultimately produces electricity.



Advantages:

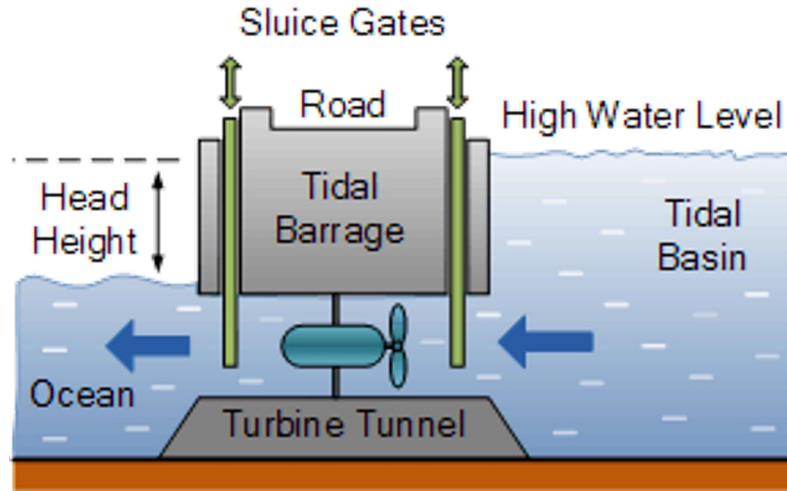
- Can be used in difficult terrain particularly willy areas for power supply
- Lese ecological impact

Disadvantages:

- Less power generation of **<25 MW**
- Only **Local** application (not to grid)
- **Seasonal** in nature (especially & non perennial rivers).
- If water NOT returned to river and used for irrigation, may lead to **water loss in lower riparian areas**

Tidal Energy

- **Mechanism:**
- High tide- Water flows from sea to tidal basin. (moves turbine to generate electricity)
- Low tide: Water goes back.
- **Requirements:**
 - Continuous high and low tide at **regular intervals**.
 - Ideally, There should be **two High and low tides** each in a day.

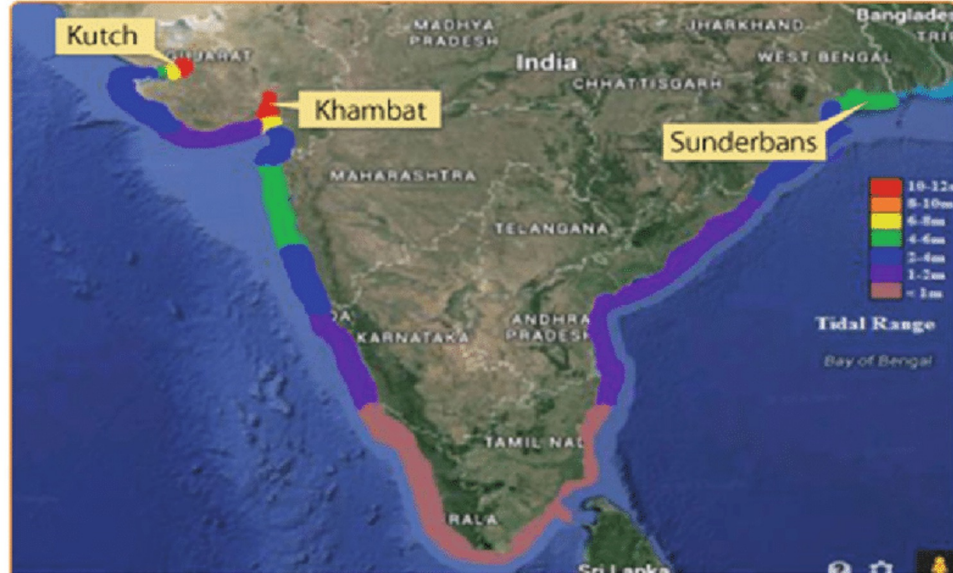


Renewable Energy



- **Tidal Range:** Minimum height difference between HT and Lt: 5 meters.
- **Locations:** Creeks, Gulfs, Estuary, Delta.
- **Locations in India:** Gulf of Kutch, Gulf of Khambat, Sundarbans Delta (Durgaduani creek project: stopped)
- **Worldwide:**
 - Only 20 locations in world where tidal power is generated.
 - **Oldest:** Rance power project (240 MW)
 - **Largest:** South Korea (Sihwa lake- 254 MW)

Renewable Energy



Gulf of Cambay/Khambhat

- Maximum Tidal Range: 11m
- Average Tidal Rang: 6.77
- Potential: 7000MW

Gulf of Kutch

- Maximum Tidal Range: 8m
- Average Tidal Rang: 5.23
- Potential: 1200MW

Sunderbans Delta

- Maximum Tidal Range: 5m
- Average Tidal Rang: 2.97
- Potential: 100MW

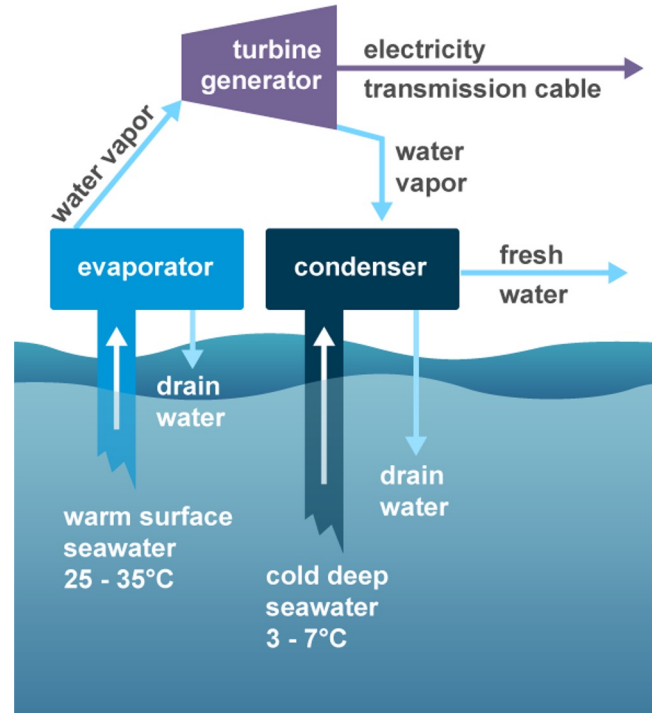
Ocean Thermal Energy Conversion (OTEC)

- **Mechanism:**

- A process for producing energy by harnessing the **temperature differences (thermal gradients)** between ocean surface waters and deep ocean waters. ○ In tropical regions, surface water, heated by Sun's energy, can be much warmer than deep water. This temperature difference can be used to **produce electricity and to desalinate ocean water.**
- OTEC systems use a temperature difference of **at least 20° Celsius** to power a turbine to produce electricity.

Renewable Energy

- Hot surface water is used to convert volatile liquid such as; Ammonia propane into gas in a heat exchanger.
- This gas turns the turbine and is condensed back to liquid – cold deep water.
- An OTEC plant can generate **100 MW power**.



Two types of OTEC technologies:

- **Closed cycle method:** A working fluid, such as **ammonia**, is **pumped through a heat exchanger and vaporized**. This vaporized steam runs a turbine. The **cold water found at the depths of the ocean condenses the vapor back to a fluid** where it returns to the heat exchanger.
 - **Open cycle system:** The **warm surface water is pressurized in a vacuum chamber and converted to steam to run the turbine**. The steam is then condensed using cold ocean water from lower depths.
-
- **India's potential:** OTEC has a theoretical potential of 180,000 MW in India subject to suitable technological evolution.

Advantages:

- 1) Carbon free
- 2) High capacity
- 3) Potential to generate potable water, hydrogen, and ammonia; Cold and nutrient rich water from the
- 4) OTEC process can benefit commercial products such as air conditioning, aquaculture, etc.

Disadvantages:

- 1) High investment
- 2) Corrosive Water
- 3) Affects marine life (**entanglement in cables, entrapment, noise generation, accidental release of biocides**)

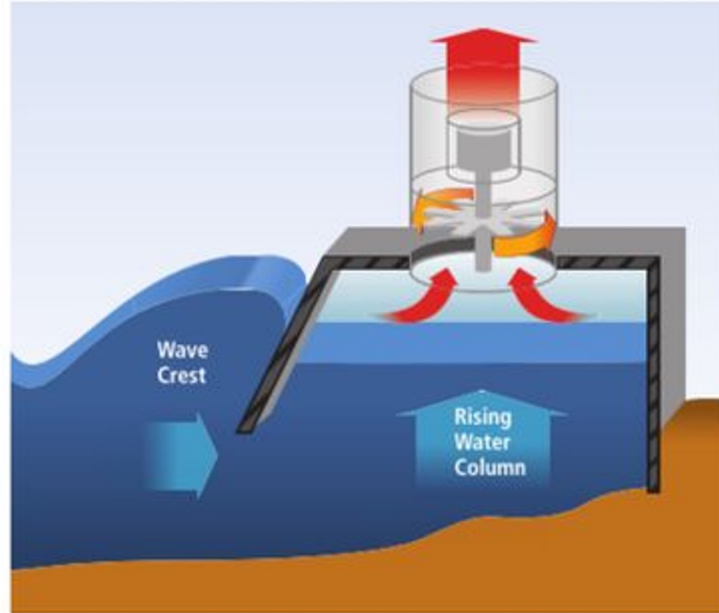
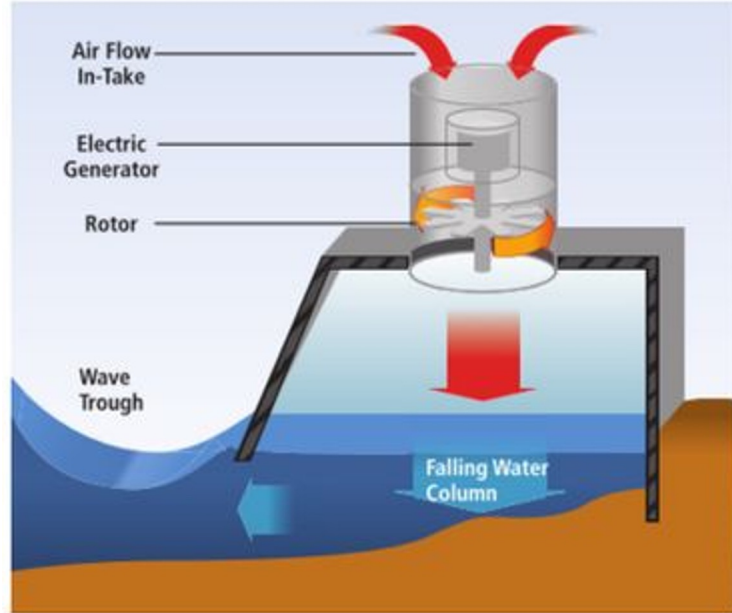
In News

- The National Institute of Ocean Technology (NIOT), an autonomous Institute under Ministry of Earth Sciences (MoES) is establishing India's first **Ocean Thermal Energy Conversion (OTEC) plant**.
- **OTEC plant will have a capacity of 65kW in Kavaratti, Lakshadweep with indigenous technology.**
 - It will power the **Low Temperature Thermal Desalination (LTTD) based** desalination plant for conversion of Sea water into Potable water.
- Earlier in 1980, India **had planned to set up an OTEC plant, off the Tamil Nadu coast, but it was abandoned.**

Sindhuja-I (Ocean Wave Energy Converter)

- It is an Ocean Wave Energy Converter developed by **IIT Madras**. It consists of a floating buoy, a spar, and an electrical module.
- Electric module generates electricity by using the relative motion between floating buoy (moving up and down with waves) and fixed spar passing through buoy centre.
- Seawater stores tidal, wave and ocean thermal energy, make the harnessing of **40 GW wave energy** possible in India.
- **Wave energy converters (WECs)** are devices that convert the kinetic and potential energy associated with a moving ocean wave into useful mechanical or electrical energy.

Renewable Energy



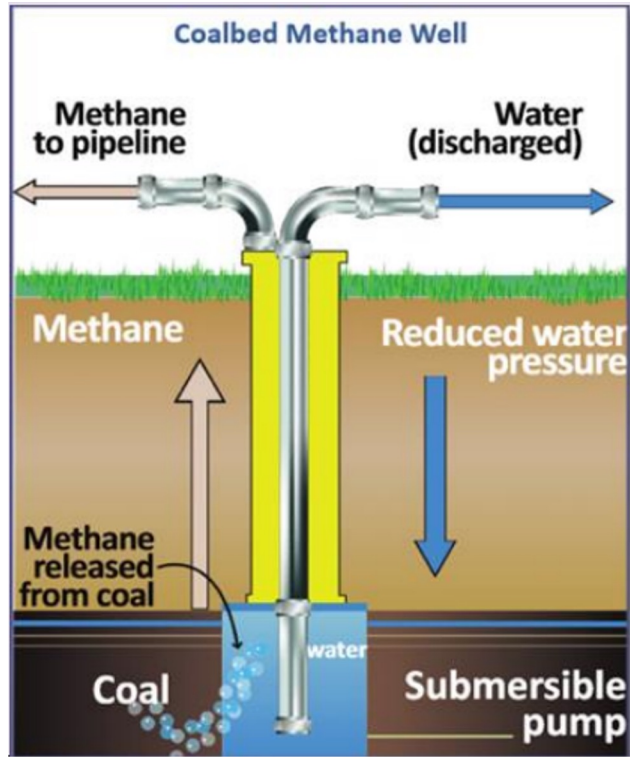
Coal-bed Methane (CBM)

- CBM is an **unconventional form of natural gas** found in coal deposits or coal seams, formed during the process of coalification, transformation of plant material into coal. When coal and methane conversion process occur such that **coal is saturated with water and methane is trapped within coal**, the result is CBM.
- **Uses:** Power generation, as CNG auto fuel, feedstock for fertilisers, industrial uses such as in cement production, rolling mills, steel plants, and for methanol production.
- Ministry of Petroleum & Natural Gas formulated CBM Policy under Oilfields (Regulation & Development) Act 1948 and Petroleum & Natural Gas Rules 1959 to harness the potential of CBM.

Process of Extraction

- A **steel-cased hole** is drilled into the coal seam and the underground (produced) water is pumped out through tubing.
- Removal of produced water **helps reduce the hydrostatic pressure within the coal bed**, causing the gas to be desorbed from its surface.
- As production occurs, the **change in pressure alters the porosity and permeability** of the coal bed.
- The recovered gas is sent into a natural gas pipeline or air compressor system.

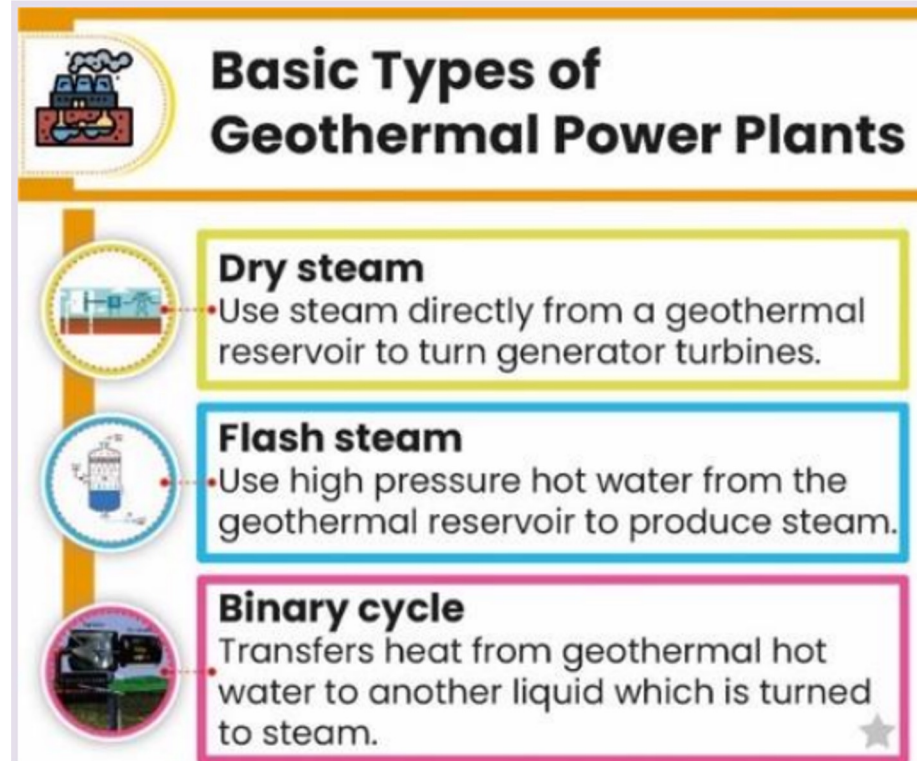
Sustainable Development



Geothermal Energy

- ONGC has reportedly started drilling its first well at Puga (Ladakh) to generate electricity using geothermal power.
- Geothermal power plants use **hydrothermal resources that have both water (hydro) and heat (thermal)**. Hot water and steam from deep underground can be piped up through underground wells and used to generate electricity in a power plant.
- **Benefits of Geothermal:** carbon-free, renewable, provides a continuous, uninterrupted supply etc.
- **Disadvantages:** Location-specific energy source, associated with other emissions like sulphur dioxide and hydrogen sulphide, causes mini tremors in area of operation, high initial cost etc.

- **Potential:** As per preliminary investigations undertaken by Geological Survey of India, there are around **300 geothermal hot springs** in India.
- **Geothermal sites (in decreasing order):**
Tattapani in Chhattisgarh, **Puga** in J&K, **Cambay** in Gujarat, **Manikaran** in Himachal Pradesh, **Surajkund** in Jharkhand, **Chumathang** in Ladakh.



● Geothermal Provinces of India

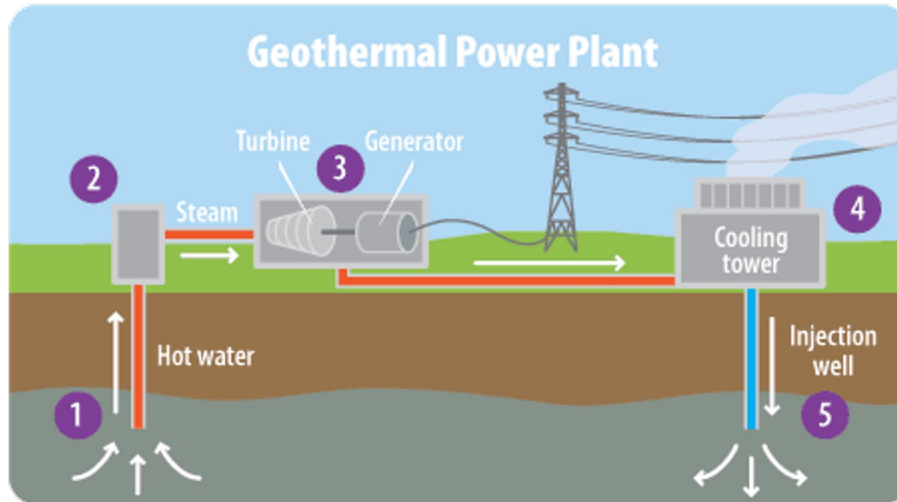
1. Himalayan belt
2. Naga-Lushai Province
3. Andaman Nicobar Islands
4. West Coast Province
5. Cambay Graben Province
6. Aravalli Province
7. Son-Narmada -Tapi province
8. Godavari Valley Province
9. Mahanadi valley Province
10. South Indian Cratonic Province

Major geothermal energy resources in India



Sustainable Development

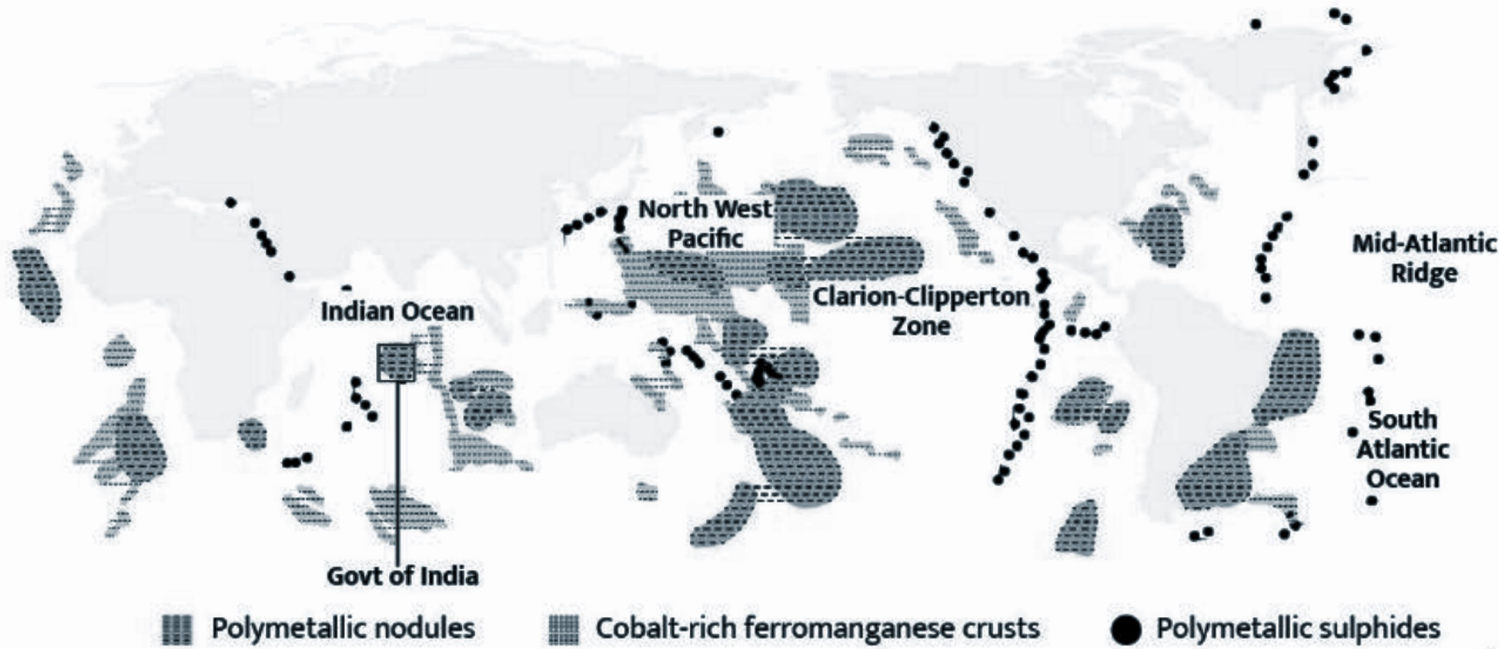
Geothermal well: used in power generation. Water circulated through a pipe in earth's crust and converted to steam. This Steam is used to run turbine for electrical power



Deep-Sea Mining

- Deep-Sea Mining involves extracting ores rich in cobalt, manganese, zinc and other rare metals from the sea floor of the deep ocean.
- India has a Deep-Sea mission for mining Central Indian Ocean Basin, access for which was granted by ISA.
- 54% of the world's ocean seafloor is beyond national jurisdiction and their mineral resources are designated as the common heritage of all mankind.

Exploration for minerals in the Area



- Deep-sea contains a number of critical minerals needed for batteries, smartphones, laptops and renewable energy capacity; found in three main resource types:
 - **Polymetallic Nodules**, i.e. the precipitated iron oxyhydroxides and manganese oxides on abyssal plain sediments at about 3,500–6,000 m water depths.
 - **Seafloor Massive Sulphides or Polymetallic Sulphides** formed along tectonic plate boundaries and volcanic provinces in water depths from <500 to 5000m.
 - **Cobalt-rich Ferromanganese Crusts** on rock outcrops on seamounts and ridges at water depths of 400–7,000 m.



International Seabed Authority (ISA)



Kingston, Jamaica



Genesis: An **autonomous international organisation** which came into existence in **1994** when **United Nations Convention on the Law of the Sea (UNCLOS), 1982** came into force.



Objective: To ensure the **effective protection of the marine environment** from harmful effects that may arise from deep-seabed related activities.



Membership: 168 members, including 167 member States and the European Union



Other key information:

UNCLOS also established 2 other bodies namely, **International Tribunal for the Law of the Sea (ITLOS)** and **Commission on the Limits of the Continental Shelf (CLCS)**.

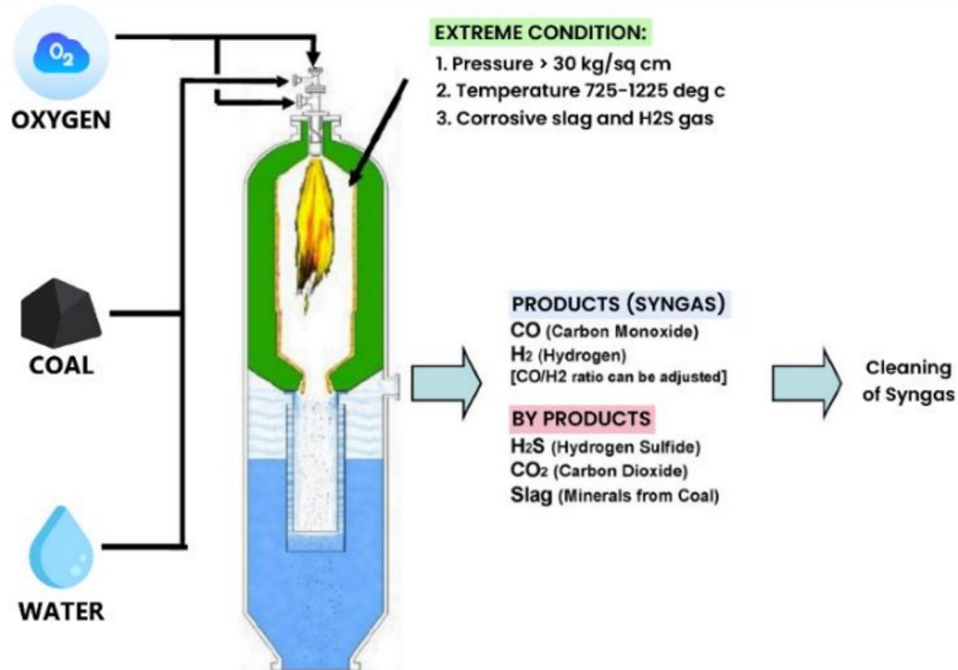
Coal Gasification

- **Context:** Union government has allowed a concession of 50% in revenue share for coal gasification.
- Coal gasification (CG) is the process **converting coal into synthesis gas** (syngas), which is a mixture of hydrogen (H₂), carbon monoxide (CO) and carbon dioxide (CO₂).

Process of Coal Gasification

- Coal is fed into a vessel called **gasifier**. In gasifier, Coal is exposed to **heat and pressure**.
- **Controlled amount of oxygen** is released into gasifier allowing the coal to burn (**Partial oxidation**)
- Heat and pressure break up the structure of partially oxidized coal causing a chemical reaction to take place creating **syn gas**.
- **Ash and other byproducts** are trapped in the bottom of the gasifier.
- In the gasification process, **sulphur** present in the coal is converted to **hydrogen sulphide (H₂S)** and minor amounts of carbonyl sulphide (COS).
 - These sulphur compounds can be easily and economically removed using **acid gas removal systems**.

BASICS OF COAL GASIFICATION PROCESS



Coal Gasification

- **In-situ gasification process:** Underground coal gasification (UCG) is an in-situ gasification process. It is an industrial process which converts coal into product gas. It carried out in non-mined coal seams using injection of oxidants and steam. The product gas is brought to the surface through production wells drilled from the surface
- **Ex-situ gasification process:** In **ex-situ process**, reactor is developed for simulating the gasification process above the surface of ground.

TYPES OF GASIFIERS

Fixed Bed Gasifiers

Flow of solids is independent of the flow of gas.
Requires minimal pre-treatment of feed coal.
Have high thermal efficiency.



Fluidized Bed Gasifiers

Characterised by uniform particle mixing, uniform temperature gradients with high char recycling rate and less capital cost.



Entrained Flow Gasifiers

Ability to handle practically any feed. Syngas obtained is free of oils and tars.



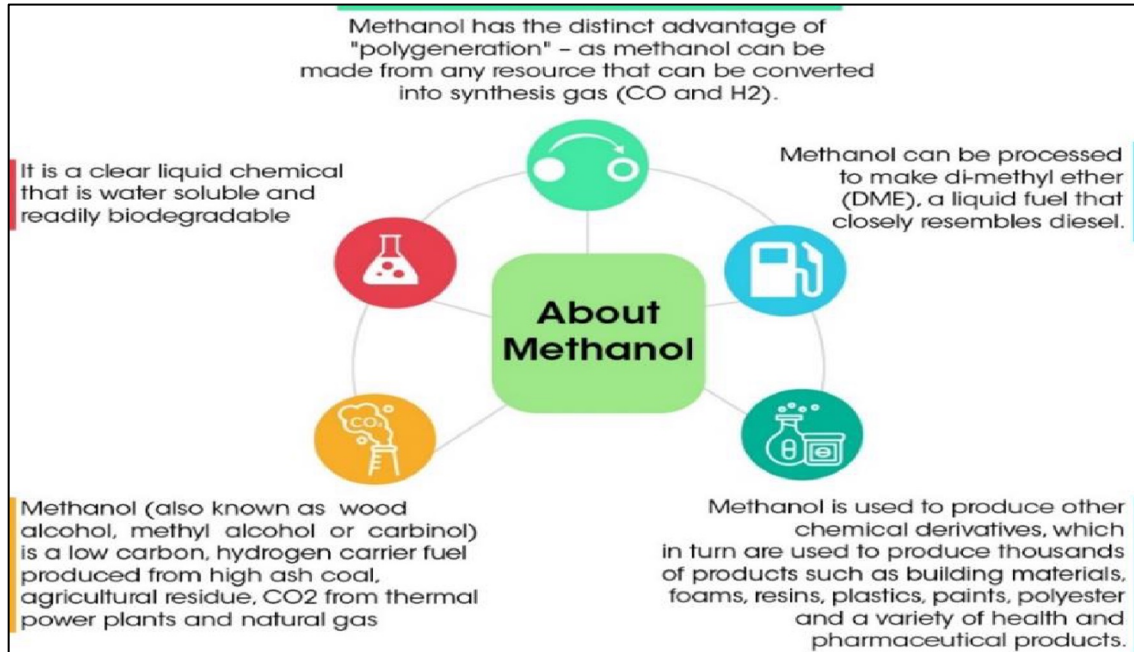
Advantages:

- **Clean** coal technology compared to burning of coal as CG plants do not produce any scrubber sludge;
- Most part of the **wash water is recycled** and the residual waste waters from gasification plants can be effectively treated.
 - Coal can be used to produce number of products using clean coal technologies
 - **Hydrogen, Methanol and Fertilizers** through coal gasification.
 - Carbon Fibres, Plastic composites using ash/ residue of the coal power plant/ gasifier.

National Coal Gasification Mission initiative

- Under Atma Nirbhar Bharat Abhiyan, **Ministry of Coal** has taken National Coal Gasification Mission initiative for utilizing coal through coal gasification and achieve 100 MT coal gasification by year 2030.
- **Nodal officer:** All coal companies have been advised to appoint a nodal officer and to prepare an action plan for **gasifying at least 10% of their coal production.**
- **Interest Rate Subvention:** Interest Rate Subvention may be provided thereby reducing the interest burden on coal gasification projects and improve the bankability of these projects.
- **Import duty exemption:** Import duty exemption for capital goods for setting up coal gasification projects may be considered.

- **Methanol Energy and Methanol Economy**



Renewable Energy



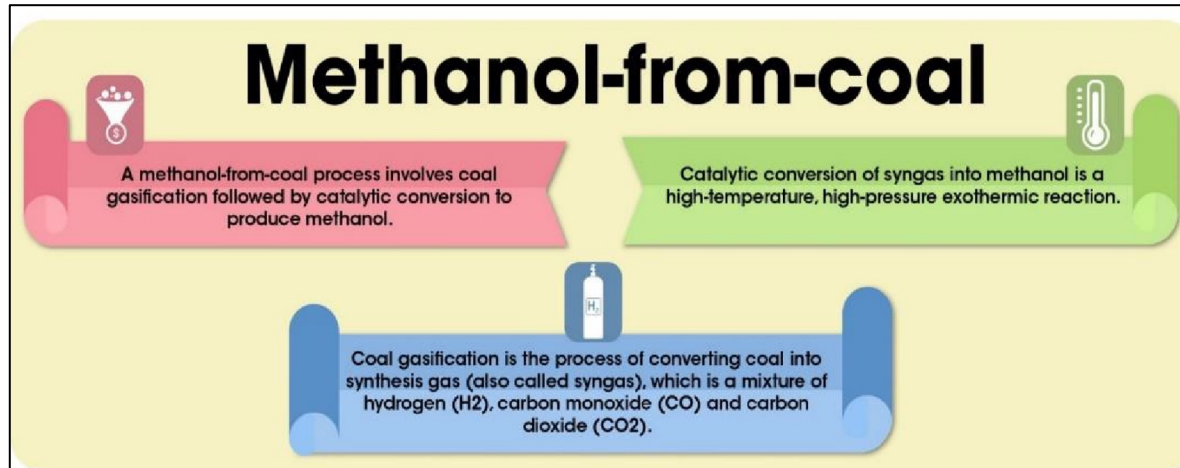
- Methanol burns efficiently in all internal combustion engines, produces no particulate matter, no soot, almost nil SO_x and NO_x emissions.
- Blending of 15% methanol in petrol will reduce pollution by 33% & diesel replacement by methanol will reduce pollution by more than 80%.
- Although slightly lower in energy content than petrol and diesel, methanol can replace both these fuels in Transport sector, Energy sector, Retail cooking etc.
- Methanol & Di-methyl Ether are substantially cheaper than Petrol and Diesel.
- Methanol has many desirable attributes which make it an excellent spark ignition engine fuel, including high octane number and improved efficiency.

Renewable Energy



- Methanol Economy is the “Bridge” to the dream of a complete “Hydrogen based fuel systems”.
- The Concept of “**Methanol Economy**” is being actively pursued by China (10% of fuel in transport sector).
- Methanol economy will help India use its vast coal reserves (burning coal directly is bad).

- Recent initiatives
- Recently, India's first Indigenously Designed **High Ash Coal Gasification Based Methanol Production Plant** was inaugurated at BHEL R&D Centre, Hyderabad.



Initiatives taken by India for Methanol based economy



NITI Aayog's road map for Methanol Economy



Bureau of Indian Standards has notified **20% DME (Di-methyl Ether, a derivative of methanol)** in LPG



Notification for **M-15, M-85, M-100 blends** issued by the Ministry of Road, Transport and Highways



In 2018, Assam Petrochemicals launched **Asia's first canister-based methanol cooking fuel programme**



BHEL, Thermax, and IIT Delhi to set up **coal-to-methanol plants** in the country

Renewable Energy



Methanol	Ethanol
Methyl Alcohol (CH_3OH)	Ethyl Alcohol (drinking alcohol) ($\text{CH}_3-\text{CH}_2-\text{OH}$)
Methanol is toxic and hence it is not suitable for consumption .	Ethanol is used in alcoholic beverages, as additive in foods, etc.
Methanol is mostly used to create other chemicals such as acetic acid and formaldehyde .	It has antiseptic properties and is found in anti-bacterial wipes and hand gels. It is also the base of many paints and perfumes because it is a good solvent.
Its odour is distinctive, and it burns as a bright white flame.	It has a strong, burning odour and will burn as a bright blue flame.
Most of the methanol is produced from coal and natural gas .	Most of the ethanol is produced by fermentation of biowaste and food crops .
• They are both used as solvents. Both are clean fuels • Both can be blended with petrol for automobile fuel.	

- **Green Methanol**
- **Context-** National Thermal Power Corporation Limited (NTPC) and Tecnimont (Italy based company) signed MoU to jointly explore green methanol production facility at a NTPC project in India.
- Green Methanol is methanol produced via a process that emits zero or a minimal amount of GHGs, usually measured as an equivalent amount of CO₂.
- It is produced from **sustainable biomass**, or from carbon dioxide and hydrogen produced from renewable electricity.
- **Applications:** material for the chemical industry, storing renewable electricity, transportation fuel and substitute fuel for maritime fuel applications

- **Methanol blended Diesel (MD15)**
- **Context:** Demo-run of Vessel powered by Methanol blended Diesel (MD15) named SB Gangadhar in Guwahati, Assam was held in run-up to India Energy Week 2023 (IEW 2023).
- IEW 2023, first major event under **India's G20 Presidency**, follows PM's pledge at COP26 to cut India's emissions to net-zero by 2070.
- IEW 2023 is held under **Ministry of Petroleum & Natural Gas**.

Renewable Energy



- **Global International Biofuel Alliance (GIBA)**
- GIBA was launched by **India during India Energy Week 2023.**
- GIBA will help create a favourable ecosystem for promoting development and deployment of biofuels with key stakeholders including US, Brazil, EU, IEA etc. as a clean mobility alternative under India's G20 Presidency.

- **Other Initiatives in news**
- **Network for Greening the Financial System**
- Reserve Bank of India (RBI) has joined the Central Banks and Supervisors Network for Greening the Financial System (NGFS) as a member.
- **About NGFS:**
- The NGFS is a group of Central banks and supervisors willing to share the best practices and contribute to the development of the environment and climate risk management in the financial sector.
- The System was launched at the **Paris One Planet Summit** in December 2017.

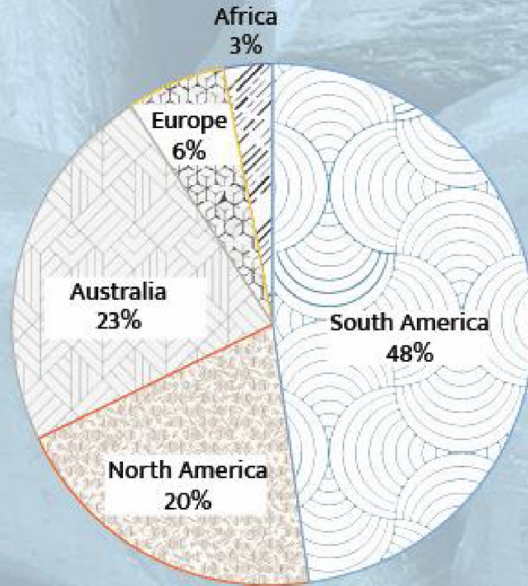
- **World's First compressed natural gas (CNG) Terminal**
- Prime Minister laid foundation stone of the World's First CNG Terminal and of the brownfield port at **Bhavnagar, Gujarat.**
- It is expected to be operational by **2026.**
- Developed using a public-private partnership (PPP) approach and has a cargo handling capacity of 1.5 million metric tonnes per annum (MMTPA).
- The upcoming project will have direct door-step connectivity to the existing roadway and railway network connecting to the largest industrial zones, Dedicated Freight Corridor and northern hinterland of India.

- **Environmental Information, Awareness, Capacity Building and Livelihood Programme (EIACP)**
- **Environmental Information System (ENVIS)** is renamed as EIACP.
- EIACP serves as a one stop platform for dissemination of environmental information, informed policy formulation on various facets of environment and facilitation of alternate livelihoods through green skilling.
- EIACP conducts **Environmental Awareness Programs** to School Children, Colleges and Universities on Environmental Important Days.
- As per the mandate of EIACP, the activities of the Programme Centres are to be aligned with **Lifestyle for the Environment (LiFE)** introduced by India at COP 26 in Glasgow.

- **Lithium-ion battery**
- **About Lithium**
- Lithium is **lightest solid metal**. It is currently produced from **hard rock** or brine mines.
- **Australia** is the world's biggest supplier, with production of Lithium from hard rock mines, while Argentina, Chile and China are producing it from salt lakes.
- **Lithium in India**
- In India first traces of Lithium ever to be discovered in the ancient igneous rock of Karnataka's **Mandya** district.
- Recently Lithium reserves were also discovered in J&K.
- Currently India **imports** all its lithium needs, with **China** and **Hong Kong** biggest suppliers.

Renewable Energy

AVAILABILITY OF LITHIUM IN THE WORLD

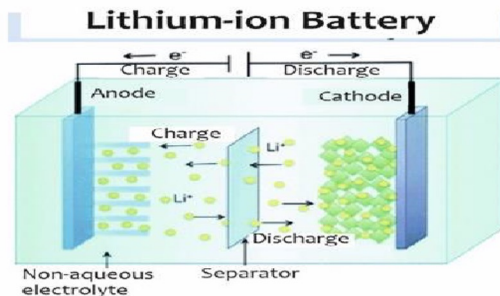




Lithium-Ion batteries

A lithium-ion battery is a family of rechargeable battery types and consists four components in a lithium-ion cell: **anode, cathode, separator, and the aqueous electrolyte.**

 Anode Graphite	 Cathode LiCoO_2
 Separator Polypropylene (PP) and and Polyethylene (PE)	 Aqueous electrolyte organic solvent and dissolved lithium salt



Advantages of Li-ion batteries

- **High specific charge density** (amount of charge a battery can supply)
- **High Voltage**
- **High energy density**
- **Low self-discharge**



Disadvantages of Li-ion batteries

- **Thermal runaway** (due to melting of the electrolyte, high operational temperature)
- **Less stability**
- **Lack of availability** for lithium raw material

- **Steps taken for Lithium-ion batteries in India**
- **National Mission on Transformative Mobility:** To encourage domestic Lithium- Ion Cell manufacturing and EV components.
- ISRO and BHEL agreement: to develop low-cost lithium-ion batteries.
- India's **first lithium cell plant manufacturing** facility will be launched in **Tirupati**, Andhra Pradesh.
- Lithium triangle nations: India is focusing on '**Lithium Triangle**' nations Argentina, Bolivia and Chile for joint exploration

Renewable Energy



Comparison of various battery types

Type	Advantage	Disadvantage
Lead Acid batteries	Mature technology, low cost and ruggedness compared with other battery technologies.	Heavy and bulky , do not cycle well.
Nickel-cadmium batteries	Availability in all sizes , can be moved around easily, can be stored, or used for up to 5 years , offer more than 1,000 charge and discharge cycles .	Environmental concerns related to cadmium. Prone to memory effect that can cause a battery to lose electrical storage area. Higher self-discharge rate than Lithium-ion.
Nickel metal hydride batteries	Widely used in consumer electronics , addresses the issues arising from the environmental effects of cadmium.	Not considered for large stationary applications because of high cost of Nickel, also have a high self-discharge rate and generally take a long time to charge.
All solid-state batteries	High energy densities and charge/discharge current limits, a higher life even in harsh conditions, lower charging time.	Challenges in both manufacturing and fundamental technology understanding .
Metal air	High energy density than lithium-ion batteries, only anode replacement needed, comparatively longer shelf life than lithium-ion.	Not electrically rechargeable , lower charge/discharge rates compared with Lithium-ion batteries
Sodium ion batteries	Wider operational temperature range and are safer. Cheaper than Lithium-ion.	Lower energy density than lithium-ion batteries.

Renewable Energy



Ecological Civilization	• Over the years, Chinese President Xi Jinping has stressed advancing ecological civilization. • Ecological Civilization describes a world in which human communities (systems of economics, agriculture, production, and consumption, etc.) are designed to promote overall well-being of people and the planet. ○ It considers nature to be part of life, rather than something that can be exploited without restraint.
Sustainable Recovery Plan	• International Energy Agency (IEA), in its latest report, has observed that carbon emissions are set to rebound as countries focus on economic recovery following the COVID-19 pandemic. • About Sustainable Recovery Plan ○ Launched in 2020, plan estimates that if governments mobilised \$1 trillion in clean energy investments each year from 2021-2023, it would boost global economic growth on an average by 1.1 percentage point a year. ○ IEA believes a full and timely implementation of the plan would save or create roughly 9 million jobs and would also help in meeting Paris Agreement goals.

Renewable Energy



Localization of Sustainable Development Goals (SDGs)	• “Localization of Sustainable Development Goals (SDGs) through Panchayati Raj Institutions (PRIs)” report was released by the Ministry of Panchayati Raj to serve as an action plan to assist the 32 lakh elected representatives of PRIs to consider for working towards localising of SDGs. • ‘Localising’ is the process of recognising subnational contexts in the achievement of the 2030 SDG agenda. • It relates both to how local and sub-national governments can support achievement of the SDGs through bottom up action as well as how SDGs can provide a framework for local development policy.
Bioeconomy	• According to recent FAO report, Renewable wood-based products, engineered wood products and wood-based textile fibres are two emerging forest product categories that can provide renewable and sustainable solutions to the global crisis. • These both products can help in realizing Bioeconomy. • Bioeconomy generally refers to an economy using renewable natural resources to produce food, energy, products, and services. • Bioeconomy covers all sectors and systems that rely on biological resources (animals, plants, micro-organisms and derived biomass, including organic waste), their functions and principles.

Reports in News



Report	Published by
Fostering Effective Energy Transition	World Economic Forum (WEF)
Environment Performance Index (EPI)	World Economic Forum (WEF) in collaboration with Yale Center for Environmental Law and Policy and Columbia University Center for International Earth Science Information Network
Renewables 2022 Global Status Report	REN21 (only global community of actors from science, governments, NGOs and industry working collectively to drive the rapid uptake of renewables)
Renewable Capacity Statistics 2022	International Renewable Energy Agency (IRENA)

Reports in News



Report	Published by
Renewable Power Generation Costs in 2021	IRENA
Renewable Energy (RE) and Jobs Annual Review 2022	IRENA and the International Labour Organization
Renewable Energy (RE) Report 2022	International Energy Association (IEA)
World Energy Employment Report 2022	International Energy Agency (IEA)
World Energy Outlook 2022	International Energy Agency (IEA) (Annual report)
Global Wind Report 2022	Global Wind Energy Council

Reports in News



Report	Published by
State of India's Environment (SoE) in figures, 2022	Centre for Science and Environment
EnviStats India 2022	National Statistics Office (NSO)
Renewables Integration in India 2021	NITI Aayog and International Energy Agency (IEA)
Transforming Food Systems for Rural Prosperity	International Fund for Agricultural Development (IFAD)
A Multi-Billion-Dollar Opportunity: Repurposing Agricultural Support to Transform Food System	FAO, UNEP, UNDP

2020

In India, why are some nuclear reactors kept “IAEA Safeguards” while others are not?

- a) Some use uranium and others use thorium
- b) Some use imported uranium and others use domestic supplies
- c) Some are operated by foreign enterprises and others are operated by domestic enterprises
- d) Some are State-owned and others are privately-owned

Answer: B) Some use imported uranium and others use domestic supplies

- There are at present 22 operational reactors, of which 14 are under the International Atomic Energy Agency (IAEA) safeguards as they use imported fuel. India currently imports uranium from Russia, Kazakhstan and Canada. Plans are also afoot to procure the fuel from Uzbekistan and Australia. The remaining "military" facilities remained off-limits to international inspectors.
- By placing the reactors under the IAEA safeguards, India gives the international nuclear energy watchdog access to them.
- This step was taken by the country in 2014 to demonstrate that its nuclear energy programme was for peaceful purposes. This is a necessary step under the Indo-US nuclear deal.

2021

Among the following which one is the least water-efficient crop?

- [A] Sugarcane
- [B] Sunflower
- [C] Pearl millet
- [D] Red gram

Answer: a) Sugarcane

- **Option a is correct.** Sugarcane is the least water efficient crop. Among the given options, its water requirement is 1800-2200 mm/season which is highest.
- **Option b is incorrect.** Sunflower's water requirement was estimated at 672.4 mm/season.
- **Option c is incorrect.** Pearl millet is a drought tolerant crop. Water requirement: 350 mm/season.
- **Option d is incorrect.** Red gram uses about 250-400 mm/season of water.

2020

According to India's National Policy on Biofuels, which of the following can be used as raw materials for the production of biofuels?

- 1) Cassava
- 2) Damaged wheat grains
- 3) Groundnut seeds
- 4) Horse gram
- 5) Rotten potatoes
- 6) Sugar beet

Select the correct answer using the code given below:

- a) 1, 2, 5 and 6 only
- b) 1, 3, 4 and 6 only
- c) 2, 3, 4 and 5 only
- d) 1, 2, 3, 4, 5 and 6

Answer: a) 1, 2, 5 and 6 only

- Under the National Policy on Biofuels ‘bioethanol’ is defined as ethanol produced from biomass such as sugar-containing materials, like sugar cane, sugar beet, sweet sorghum, etc.; starch-containing materials such as corn, cassava, rotten potatoes, algae, etc.; and, cellulosic materials such as bagasse, wood waste, agricultural and forestry residues or other renewable resources like industrial waste.
- For Ethanol Production the following raw materials may be potentially used: B-Molasses, Sugarcane juice, biomass in form of grasses, agriculture residues (Rice straw, cotton stalk, corn cobs, sawdust, bagasse, etc.), sugar-containing materials like sugar beet, sweet sorghum, etc. and starch containing materials such as corn, cassava, rotten potatoes, etc., Damaged food grains like wheat, broken rice, etc. which are unfit for human consumption, Food grains during the surplus phase. Algal feedstock and the cultivation of seaweeds can also be a potential feedstock for ethanol production. For Biodiesel Production: Nonedible Oilseeds, Used Cooking Oil (UCO), Animal tallow, Acid Oil, Algal feedstock, etc. For Advanced Biofuels: Biomass, MSW, Industrial waste, Plastic waste, etc.

2017

It is possible to produce algae based biofuels, but what is/are the likely limitation(s) of developing countries in promoting this industry?

1. Production of algae based biofuels is possible in seas only and not on continents.
2. Setting up and engineering the algae based biofuel production requires high level of expertise/technology until the construction is completed.
3. Economically viable production necessitates the setting up of large scale facilities which may raise ecological and social concerns.

Select the correct answer using the code given below:

- a) 1 and 2 only
- b) 2 and 3 only
- c) 3 only
- d) 1, 2 and 3

Answer- B) 2 and 3 only

- Algae can be grown using land and water not suitable for plant and food production. The algal biofuel technology includes selection of specific species for production and extraction of valuable co-products.
- Algae is bioengineered for achieving advanced photosynthetic efficiencies through continued development of production system. To make the industry commercially viable, researchers have pursued biotech varieties, which could be particularly dangerous if released into nature.
- Some algae biodiesel proposals involve aquaculture-style operations in open ocean waters, which could have harmful ecological effects, especially if biotech algae is used. Water and nutrient use would also be extreme, making any serious scaling up of algae biofuels quite unsustainable.

2013

Consider the following organisms:

1. Agaricus
2. Nostoc
3. Spirogyra

Which of the above is/are used as biofertilizer/biofertilizers

- a) 1 and 2
- b) 2 only
- c) 2 and 3
- d) 3 only

Answer: B

- Biofertilizers are products containing carrier-based (solid or liquid) living microorganisms which are agriculturally useful in terms of nitrogen fixation, phosphorus solubilization, or nutrient mobilization to increase the productivity of soil or crop.
- Classification of biofertilizers:
 - Bacterial Biofertilizers: Rhizobium, Azospirillum, Azotobacter, Phosphobacteria, Nostoc, etc.
- Agaricus is an **edible fungus** and is commonly known as a mushroom.
- Spirogyra is a **large genus of freshwater** green algae found in shallow ponds, ditches, and amongst vegetation at the edges of large lakes, generally growing free-floating. **Hence, option b is correct.**

2013

With reference to the usefulness of the by-products of the sugar industry, which of the following statements is/are correct?

1. Bagasse can be used as biomass fuel for the generation of energy.
2. Molasses can be used as one of the feedstocks for the production of synthetic chemical fertilizers.
3. Molasses can be used for the production of ethanol.

Select the correct answer using the codes given below.

- a) 1 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

Answer: C

- Bagasse is a remnant left after extracting the juice of the sugarcane or sorghum known as bagasse. It can be used as a fuel for the generation of heat energy and electricity, in the production of pulp, and paper and to feed cattle. **Hence, statement 1 is correct.**
- Molasses is the thick syrup that results from the refining of sugarcane or sugar beets into sugar. It can be used as sweetening and flavoring foods, in the production of commercial brown sugar, baking, ethanol, and cattle feed. **Hence, statement 3 is correct.**

2014

With reference to two non-conventional energy sources called ‘coal bed methane’ and ‘shale gas’, consider the following ‘statements:

1. Coal bed methane is the pure methane gas extracted from coal seams, while shale gas is a mixture of propane and butane only that can be extracted from fine-grained sedimentary rocks.
2. In India abundant coal bed methane sources exist, but so far no shale gas sources have been found.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Answer: D

- Coalbed methane is a non-conventional and is a major source of natural gas. It is extracted from coal seams.
- Shale gas is a natural gas and is a hydrocarbon gas mixture. It consists of methane and other hydrocarbons like ethane, propane, butane, carbon dioxide, nitrogen, and hydrogen sulfide. **Hence, statement 1 is incorrect.**
- India has identified six basins of shale gas; Cambay (Gujarat), Assam-Arakan (North East), Gondwana (Central India), Krishna Godavari onshore (East Coast), Cauvery Onshore, and Indo-Gangetic basins. **Hence, statement 2 is incorrect.**

2014

In India, cluster bean (Guar) is traditionally used as a vegetable or animal feed, but recently the cultivation of this has assumed significance. Which one of the following statements is correct in this context?

- a) The oil extracted from seeds is used in the manufacture of biodegradable plastics
- b) The gum made from its seeds is used in the extraction of shale gas
- c) The leaf extract of this plant has the properties of antihistamines
- d) It is a source of high-quality biodiesel

Answer: B

- The **guar or cluster bean**, with the **botanical name Cyamopsis tetragonoloba**, is an annual legume and the source of guar gum.
- It is also known as **Gavar, guar, or guvar bean**.
- The gum made from its seeds is used in the **extraction of shale gas**
- The seeds of the guar bean contain a large endosperm.
- This endosperm consists of a large polysaccharide of galactose and mannose.
- This polymer is water-soluble and exhibits a viscosifying effect in the water.
- Guar gum has a multitude of different applications in food products, industrial products, and the extractive industry.

2019

In the context of proposals for the use of hydrogen-enriched CNG (H-CNG) as fuel for buses in public transport, consider the following statements:

1. The main advantage of the use of H-CNG is the elimination of carbon monoxide emissions.
2. H-CNG as fuel reduces carbon dioxide and hydrocarbon emissions.
3. Hydrogen up to one-fifth by volume can be blended with CNG as fuel for buses.
4. H-CNG makes the fuel less expensive than CNG.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 and 3 only
- c) 4 only
- d) 1, 2, 3 and 4

Answer) B

- H -CNG is a blend of hydrogen and CNG, the ideal hydrogen concentration being 18%. Compared to conventional CNG, the use of H - CNG can reduce the emission of carbon monoxide by up to 70% and 15% reduction in total hydrocarbon emissions, besides enabling up to 5% savings in fuel.
- In its report to the Supreme Court, the EPCA has estimated that to fuel Delhi's 5,500 buses, about 400 tonnes H -CNG would be needed per day. Setting up four fuel-dispensing facilities would cost Rs 330 crore, which can be funded from the Environment Compensation Charge (ECC) fund made up of cess on commercial vehicles entering Delhi, it said. For consumers who pay Rs 42 per kg for CNG, the cost of H -CNG would not be more than Rs 43 per kg.

2022

System of Rice Intensification' of cultivation, in which alternate wetting and drying of rice fields is practised, results in :

1. Reduced seed requirement
2. Reduced methane production
3. Reduced electricity consumption

Select the correct answer using the code given below :

- [A] 1 and 2 only
- [B] 2 and 3 only
- [C] 1 and 3 only
- [D] 1, 2 and 3

Answer- D

- System of Rice Intensification (SRI) includes a method called Alternate Wetting and Drying (AWD) which is a form of controlled or intermittent irrigation of the rice crops.
- SRI methods have the following benefits and impacts, in general, compared to conventional methods of paddy cultivation:
 - To increase paddy yields usually by 20-50% and sometimes up to 100%
 - It reduces required seeds for transplanting by 60-80%(Hence Option 1 is correct);

- The electricity consumption will be less as it enables reduce use of chemical fertilizers and agrichemicals and reduced use of irrigation water by 25-50%
(Hence Option 3 is correct)
- **Option 2 is correct.** By stopping flooding and relying mostly on organic fertilization, methane emissions are greatly reduced without offsetting increases in nitrous oxide. Also, less carbon dioxide is generated due to reduced reliance on and transport of manufactured inputs. Oxford researchers in India found 30% reduction in net greenhouse gas emissions with SRI, compared with current practice, and more than 50% reduction in greenhouse gas emissions per kilogram of rice produced.

2012

To meet its rapidly growing energy demand, some opine that India should pursue research and development on Thorium as the future fuel of nuclear energy. In this context, what advantage does Thorium hold over uranium?

1. Thorium is far more abundant in nature than uranium
2. On the basis of per unit mass of mined mineral, thorium can generate more energy compared to natural uranium.
3. Thorium produces less harmful waste in comparison to uranium.

Which of the statements given above is/are Correct?

- a) 1 Only
- b) 2 & 3 Only
- c) 1 & 3 Only
- d) 1,2 & 3

Answer- D

- Thorium is 3 times as abundant as uranium
- It is estimated that 1 tonne of thorium can produce as much energy as 35 tons of uranium
- Thorium reactors are more efficient than uranium reactors, generating less waste.

2016

India is an important member of the ‘International Thermonuclear Experimental Reactor’. If this experiment succeeds, what is the immediate advantage for India?

- (a) It can use thorium in place of uranium for power generation
- (b) It can attain a global role in satellite navigation
- (c) It can drastically improve the efficiency of its fission reactors in power generation
- (d) It can build fusion reactors for power generation

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2022

With reference to India, consider the following statements:

1. Monazite is a source of rare earths.
2. Monazite contains thorium.
3. Monazite occurs naturally in the entire Indian coastal sands in India.
4. In India, Government bodies only can process or export monazite.

Which of the statements given above are correct?

- a) 1, 2 and 3 only
- b) 1, 2 and 4 only
- c) 3 and 4 only
- d) 1, 2, 3 and 4

Answer: B

- **Statement 1 and 2 are correct.**
 - Monazite is a major source of rare earths and thorium.
 - It yields a number of rare-earth elements, such as neodymium and praseodymium. Both of these are in demand internationally for making high-performance rare-earth magnets (components of power wind turbines, electric vehicles and robotics)

- **Statement 3 is incorrect.** Though monazite is found in most coastal areas of India. It is questionable to be found along entire coast. The main mines are found along the coasts of southern India in Kerala, Tamil Nadu and in Orissa.
- **Statement 4 is correct.** Export of monazite without a license from AERB is a violation of the Atomic Energy (Radiation Protection) Rules 2004. Indian Rare Earths Limited (IREL), a wholly owned Public Sector Undertaking of the Government of India (GOI) under DAE, is the only entity which has been permitted to produce and process monazite, and handle it for domestic use as well as for export.